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Electrode assembly, battery, and battery pack and vehicle including the same

Abstract

An electrode assembly, a battery, and a battery pack and a vehicle including the same are provided. In the electrode assembly, a first electrode, a second electrode, and a separator interposed therebetween are wound based on an axis to define a core and an outer circumference. The first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly. A part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region including stacked layers of the second portion, and in a partial region of the first surface region, a number of the stacked layers of the second portion is 10 or more in the in the first axial direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This non-provisional application claims the benefit under 35 U.S.C. § 119(a) to Patent Application No. 10-2021-0022881, filed in the Republic of Korea on Feb. 19, 2021, Patent Application No. 10-2021-0022891, filed in the Republic of Korea on Feb. 19, 2021, Patent Application No. 10-2021-0022894, filed in the Republic of Korea on Feb. 19, 2021, Patent Application No. 10-2021-0022897, filed in the Republic of Korea on Feb. 19, 2021, Patent Application No. 10-2021-0024424, filed in the Republic of Korea on Feb. 23, 2021, Patent Application No. 10-2021-0030291, filed in the Republic of Korea on Mar. 8, 2021, Patent Application No. 10-2021-0030300, filed in the Republic of Korea on Mar. 8, 2021, Patent Application No. 10-2021-0046798, filed in the Republic of Korea on Apr. 9, 2021, Patent Application No. 10-2021-0058183, filed in the Republic of Korea on May 4, 2021, Patent Application No. 10-2021-0077046, filed in the Republic of Korea on Jun. 14, 2021, Patent Application No. 10-2021-0084326, filed in the Republic of Korea on Jun. 28, 2021, Patent Application No. 10-2021-0131205, filed in the Republic of Korea on Oct. 1, 2021, Patent

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(2) Also, Patent Application No. 10-2021-0007278, filed in the Republic of Korea on Jan. 19, 2021, is hereby expressly incorporated by reference in its entirety into the present application.

TECHNICAL FIELD

(3) The present disclosure relates to an electrode assembly, a battery, and a battery pack and a vehicle including the same.

BACKGROUND ART

(4) Secondary batteries that are easily applicable to various product groups and have electrical characteristics such as high energy density are universally applied not only to portable devices but also to electric vehicles (EVs) or hybrid electric vehicles (HEVs) driven by an electric drive source.

(5) These secondary batteries are attracting attention as a new energy source to improve eco-friendliness and energy efficiency because they have the primary advantage that they can dramatically reduce the use of fossil fuels as well as the secondary advantage that no by-products are generated from the use of energy.

(6) Secondary batteries currently widely used in the art include lithium ion batteries, lithium polymer batteries, nickel cadmium batteries, nickel hydrogen batteries, nickel zinc batteries, and the like. A unit secondary battery, namely a unit battery, has an operating voltage of about 2.5V to 4.5V. Therefore, when a higher output voltage is required, a battery pack may be configured by connecting a plurality of batteries in series. In addition, a plurality of batteries may be connected in parallel to form a battery pack according to the charge/discharge capacity required for the battery pack. Accordingly, the number of batteries included in the battery pack and the form of electrical connection may be variously set according to the required output voltage and/or charge/discharge capacity.

(7) Meanwhile, as a kind of unit secondary battery, there are known cylindrical, rectangular, and pouch-type batteries. In the case of a cylindrical battery, a separator serving as an insulator is interposed between a positive electrode and a negative electrode, and they are wound to form an electrode assembly in the form of a jelly roll, which is inserted into a battery housing to configure a battery. In addition, a strip-shaped electrode tab may be connected to an uncoated portion of each of the positive electrode and the negative electrode, and the electrode tab electrically connects the electrode assembly and an electrode terminal exposed to the outside. For reference, the positive electrode terminal is a cap of a sealing body that seals the opening of the battery housing, and the

negative electrode terminal is the battery housing. However, according to the conventional cylindrical battery having such a structure, since current is concentrated in the strip-shaped electrode tab coupled to the uncoated portion of the positive electrode and/or the uncoated portion of the negative electrode, the current collection efficiency is not good due to large resistance and large heat generation.

(8) For small cylindrical batteries with a form factor 1865 (diameter: 18 mm, height: 65 mm) or a form factor 2170 (diameter: 21 mm, height: 70 mm), resistance and heat are not a major issue. However, when the form factor is increased to apply the cylindrical battery to an electric vehicle, the cylindrical battery may ignite while a lot of heat is generated around the electrode tab during the rapid charging process.

(9) In order to solve this problem, there is provided a cylindrical battery (so-called tab-less cylindrical battery) in which the uncoated portion of the positive electrode and the uncoated portion of the negative electrode are designed to be positioned at the top and bottom of the jelly-roll type electrode assembly, respectively, and the current collector is welded to the uncoated portion to improve the current collecting efficiency.

(10) FIGS. 1 to 3 are diagrams showing a process of manufacturing a tab-less cylindrical battery. FIG. 1 shows the structure of an electrode, FIG. 2 shows a process of winding the electrode, and FIG. 3 shows a process of welding a current collector to a bending surface region of an uncoated portion.

(11) Referring to FIGS. 1 to 3, a positive electrode **10** and a negative electrode **11** have a structure in which a sheet-shaped current collector **20** is coated with an active material **21**, and include an uncoated portion **22** at one long side along the winding direction X.

(12) An electrode assembly A is manufactured by sequentially stacking the positive electrode **10** and the negative electrode **11** together with two sheets of separators **12** as shown in FIG. 2 and then winding them in one direction X. At this time, the uncoated portions of the positive electrode **10** and the negative electrode **11** are arranged in opposite directions. The positions of the positive electrode **10** and the negative electrode **11** may be changed opposite to those shown in the figures.

(13) After the winding process, the uncoated portion **10a** of the positive electrode **10** and the uncoated portion **11a** of the negative electrode **11** are bent toward the core to form a bending surface region. After that, current collectors **30**, **31** are welded and coupled to the uncoated portions **10a**, **11a**, respectively.

(14) An electrode tab is not separately coupled to the positive electrode uncoated portion **10a** and the negative electrode uncoated portion **11a**, the current collectors **30**, **31** are connected to external electrode terminals, and a current path is formed with a large cross-sectional area along the winding axis direction of electrode assembly A (see arrow, which has an advantage of lowering the resistance of the battery. This is because resistance is inversely proportional to the cross-sectional area of the path through which the current flows.

(15) In the tab-less cylindrical battery, in order to improve the welding characteristics between the uncoated portions **10a**, **11a** and the current collectors **30**, **31**, a strong pressure should be applied to the welding regions of the uncoated portions **10a**, **11a** to bend the uncoated portions **10a**, **11a** as flat as possible.

(16) When the uncoated portions **10a**, **11a** are bent, as the uncoated portion **32** adjacent to the core of the electrode assembly A is bent, all or a significant portion of the cavity **33** in the core of the electrode assembly A is blocked. In this case, it causes a problem in the electrolyte injection process. That is, the cavity **33** in the core of the electrode assembly A is used as a passage through which an electrolyte is injected. However, if the corresponding passage is blocked, electrolyte injection is difficult. In addition, while an electrolyte injector is being inserted into the cavity **33**, the electrolyte injector may interfere with the uncoated portion **32** bent near the core, which may cause the uncoated portion **32** to tear.

(17) In addition, the bent portions of the uncoated portions **10a**, **11a** to which the current collectors

30, 31 are welded should be overlapped in multiple layers and there should not be any empty spaces (gaps). In this way, sufficient welding strength may be obtained, and even with the latest technology such as laser welding, it is possible to prevent laser from penetrating into the electrode assembly A and melting the separator or the active material.

(18) In order for the uncoated portions **10a, 11a** to be overlapped with the same number of layers, the uncoated portions **10a, 11a** at the corresponding positions based on the position of each winding turn should be bent toward the core and cover the top surface of the uncoated portion bent at an inner winding turn. In addition, assuming that the interval between winding turns is d and the bending length of the uncoated portions **10a, 11a** of each winding turn is e , the bending length e should have a length greater than $d \cdot n$ (n is a natural number greater than or equal to 2). Only in this case, an area where the uncoated portions **10a, 11a** are overlapped in multiple layers with the same amount is formed. In addition, in order to sufficiently obtain a region in which the uncoated portions **10a, 11a** are overlapped in substantially the same number in the radial direction of the electrode assembly, the uncoated portions **10a, 11a** should have a sufficient length. However, since the electrode assembly included in a small cylindrical battery has a small radius, it is difficult to conceive of a motivation for deriving the concept of designing the uncoated portions **10a, 11a** having a sufficiently long bending length.

DISCLOSURE

Technical Problem

(19) The present disclosure is designed to solve the problems of the related art, and therefore the present disclosure is directed to providing an electrode assembly having an uncoated portion bent structure that may prevent a separator or an active material layer from being damaged when welding a current collector by sufficiently securing a region in which uncoated portions are overlapped into 10 or more layers in a radial direction of an electrode assembly when the uncoated portions exposed at both ends of the electrode assembly are bent.

(20) The present disclosure is also directed to providing an electrode assembly in which an electrolyte injection passage is not blocked even if the uncoated portion is bent.

(21) The present disclosure is also directed to providing an electrode assembly with improved energy density and reduced resistance.

(22) The present disclosure is also directed to providing a battery including the electrode assembly having an improved structure, a battery pack including the battery, and a vehicle including the battery pack.

(23) The technical objects to be solved by the present disclosure are not limited to the above, and other objects not mentioned herein will be clearly understood by those skilled in the art from the following disclosure.

Technical Solution

(24) In one aspect of the present disclosure, there is provided an electrode assembly including a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference of the electrode assembly, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region including stacked layers of the second portion, and wherein, in a partial region of the first surface region, a number of the stacked layers of the second portion may be 10 or more in the first axial direction.

(25) A number of total winding turns of the first electrode may be defined as $n_{sub.1}$, a relative radial position $R_{sub.1,k}$ may be defined by dividing a winding turn index k at a $k_{sup.th}$ winding turn location of the first electrode by the number of total winding turns $n_{sub.1}$, k being a natural number of 1 to $n_{sub.1}$, and a ratio between a length of a radial region including the relative radial

position $R_{sub.1,k}$ having 10 or more stacked layers of the second portion and a length of the first surface region may be 30% or more.

(26) The ratio may be 30% to 85%.

(27) The second electrode may include a third portion coated with an active material and a fourth portion at a second side and adjacent the third portion, the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, a part of the fourth portion may be bent in the radial direction of the electrode assembly forming a second surface region including stacked layers of the fourth portion, and, in a partial region of the second surface region, a number of the stacked layers of the fourth portion may be 10 or more in the second axial direction of the electrode assembly.

(28) A number of total winding turns of the second electrode may be defined as $n_{sub.2}$, a relative radial position $R_{sub.2,k}$ may be defined by dividing a winding turn index k at a $k_{sup.th}$ winding turn location of the second electrode by the number of total winding turns $n_{sub.2}$, k being a natural number of 1 to $n_{sub.2}$, and a ratio between a length of a radial region including the relative radial position $R_{sub.2,k}$ having 10 or more stacked layers of the fourth portion and a length of the second surface region may be 30% or more

(29) The ratio may be 30% to 85%.

(30) A region of the second portion from a relative radial position $R_{sub.1,1}$ of a $1_{sup.st}$ winding turn of the first electrode to a relative radial position $R_{sub.1,k^*}$ of a $k^*_{sup.th}$ winding turn of the first electrode may have a smaller height than a region of the second portion from a relative radial position $R_{sub.1,k^*+1}$ of a $k^*+1_{sup.th}$ winding turn of the first electrode to a relative radial position of a $n_{sub.1}_{sup.th}$ winding turn of the first electrode.

(31) A region of the second portion from a relative radial position $R_{sub.1,1}$ of $1_{sup.st}$ winding turn of the first electrode to a relative radial position $R_{sub.1,k^*}$ of a $k^*_{sup.th}$ winding turn of the first electrode may have a smaller height than the first surface region.

(32) A region of the second portion from a relative radial position $R_{sub.1,1}$ of a $1_{sup.st}$ winding turn of the first electrode to a relative radial position $R_{sub.1,k^*}$ of a $k^*_{sup.th}$ winding turn may include one or more layers of the second portion not bent toward the core of the electrode assembly.

(33) A region of the fourth portion from a relative radial position $R_{sub.2,1}$ of $1_{sup.st}$ winding turn of the second electrode to a relative radial position $R_{sub.2,k^*}$ of a $k^*_{sup.th}$ winding turn of the second electrode may have a smaller height than a region of the fourth portion from a relative radial position $R_{sub.2,k^*+1}$ of a $k^*+1_{sup.th}$ winding turn of the second electrode to a relative radial position of a $n_{sub.2}_{sup.th}$ winding turn of the second electrode.

(34) A region of the fourth portion from a relative radial position $R_{sub.2,1}$ of a $1_{sup.st}$ winding turn of the second electrode to a relative radial position $R_{sub.2,k^*}$ of a $k^*_{sup.th}$ winding turn of the second electrode may have a smaller height than the second surface region.

(35) A region of the fourth portion from a relative radial position $R_{sub.2,1}$ of a 1st winding turn of the second electrode to a relative radial position $R_{sub.2,k^*}$ of a $k^*_{sup.th}$ winding turn of the second electrode may include one or more layers of the fourth portion not bent toward the core of the electrode assembly.

(36) The second portion or the fourth portion may include a plurality of segments, and each of the plurality of segments may be bendable.

(37) Each of the plurality of segments may have a geometric shape including a base at a bending line of each of the plurality of the segments, and the geometric shape may include one or more straight lines, one or more curves, or a combination thereof.

(38) The geometric shape may have a width decreasing stepwise or gradually from the base to a top of the geometric shape.

(39) A lower internal angle of the geometric shape between the base and a side of at least one of the plurality of segments intersecting the base may be 60 degrees to 85 degrees.

(40) The plurality of segments may have a plurality of lower internal angles, and the plurality of lower internal angles may increase stepwise or gradually along a direction parallel to a winding direction of the electrode assembly.

(41) Each of the plurality of segments may have a geometric shape having a base at a bending line of each of the plurality of segments, and a lower internal angle $\theta_{\text{assumption}}$ and a lower internal angle $\theta_{\text{sub.real}}$ of each of the plurality of segments may satisfy the following formula:

$\theta_{\text{sub.real}} > \theta_{\text{sub.assumption}}$; and

$\theta_{\text{sub.assumption}} = 90^\circ - 360^\circ \cdot (L_{\text{sub.arc}} / 2\pi r) \cdot 0.5$, wherein $\theta_{\text{sub.assumption}}$ may be a lower internal angle of each of the plurality of segments at a winding turn based on adjacent sides of a pair of the plurality of segments being parallel, wherein $\theta_{\text{sub.real}}$ may be an actual lower internal angle at the winding turn, wherein r may be a radius of the winding turn from a core center of the electrode assembly to each of the plurality of segments at the winding turn, and wherein $L_{\text{sub.arc}}$ may be an arc length corresponding to a lower portion of each of the plurality of segments at the winding turn.

(42) A circumferential angle corresponding to the $L_{\text{sub.arc}}$ may be 45 degrees or less.

(43) An overlapping ratio of the adjacent sides of the pair of the plurality of segments at the winding turn may satisfy the following formula:

$\theta_{\text{sub.real}} / \theta_{\text{sub.assumption}} - 1$, wherein the overlapping ratio of the pair of the plurality of segments may be greater than 0 and less than 0.05.

(44) A virtual circle passing through an adjacent pair of the plurality of segments at a winding turn having a radius from a core center of the electrode may include a pair of arcs passing through each segment of the adjacent pair of the plurality of segments overlapping with each other.

(45) A ratio between a length of an overlapping arc of the pair of arcs to a length of each arc of the pair of arcs passing through each of the pair of the plurality segments may be defined as an overlapping ratio, and the overlapping ratio may be greater than 0 and less than 0.05.

(46) A region of the second portion from a relative radial position $R_{\text{sub.1,1}}$ of a $1_{\text{sup.st}}$ winding turn of the first electrode to a relative radial position $R_{\text{sub.1,k}^*}$ of a $k^*_{\text{sup.th}}$ winding turn of the first electrode may have a smaller height than a region of the second portion from a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^*+1_{\text{sup.th}}$ winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1,1}}_{\text{sup.th}}$ winding turn of the first electrode, and the region from the relative radial position $R_{\text{sub.1,1}}$ of the $1_{\text{sup.st}}$ winding turn to the relative radial position $R_{\text{sub.1,k}^*}$ of the $k^*_{\text{sup.th}}$ winding turn may not be bent toward the core.

(47) A length of the first electrode corresponding to the region from the relative radial position $R_{\text{sub.1,1}}$ to the relative radial position $R_{\text{sub.1,k}^*}$ may be 1% to 30% compared to a length of the first electrode corresponding to the region from the relative radial position $R_{\text{sub.1,k}^*+1}$ to the relative radial position of the $n_{\text{sub.1,1}}_{\text{sup.th}}$ winding turn of the first electrode.

(48) A bending length $fd_{\text{sub.1,k}^*+1}$ of the second portion at a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^*+1_{\text{sup.th}}$ winding turn of the first electrode may be shorter than a radial length from a relative radial position $R_{\text{sub.1,1}}$ of $1_{\text{sup.st}}$ winding turn of the first electrode to a relative radial position $R_{\text{sub.1,k}^*}$ of a $k^*_{\text{sup.th}}$ winding turn of the first electrode.

(49) A radius of the core of the electrode assembly may be defined as $r_{\text{sub.c}}$, and a region from a center of the core to $0.90r_{\text{sub.c}}$ may not be blocked by a bent portion of the second portion located in a region from a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^*+1_{\text{sup.th}}$ winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1,1}}_{\text{sup.th}}$ winding turn of the first electrode.

(50) A bending length $fd_{\text{sub.1,k}^*+1}$ of the second portion at the relative radial position $R_{\text{sub.1,k}^*+1}$ of the $k^*+1_{\text{sup.th}}$ winding turn, the radius $r_{\text{sub.c}}$ of the core, and a distance $d_{\text{sub.1,k}^*+1}$ from a center of the electrode assembly to the relative radial position $R_{\text{sub.1,k}^*+1}$ may satisfy the following formula:

$fd_{\text{sub.1,k}^*+1} + 0.90 \cdot r_{\text{sub.c}} \leq d_{\text{sub.1,k}^*+1}$.

(51) A region of the fourth portion from a relative radial position $R_{\text{sub.2,1}}$ of a $1_{\text{sup.st}}$ winding

turn of the second electrode to a relative radial position $R_{\text{sub.2},k^*}$ of a $k^{\text{sup.th}}$ winding turn may have a smaller height than a region of the fourth portion from a relative radial position $R_{\text{sub.2},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn of the second electrode to a relative radial position of $n.\text{sub.2}.\text{sup.th}$ winding turn of the second electrode, and the region from the relative radial position $R_{\text{sub.2},1}$ of the $1.\text{sup.st}$ winding turn to the relative radial position $R_{\text{sub.2},k^*}$ of a $k^{\text{sup.th}}$ winding turn may not be bent toward the core.

(52) A length of the second electrode corresponding to the region from the relative radial position $R_{\text{sub.2},1}$ to the relative radial position $R_{\text{sub.2},k^*}$ may be 1% to 30% compared to a length of the second electrode corresponding to the region from the relative radial position $R_{\text{sub.2},k^*+1}$ to the relative radial position of $n.\text{sub.2}.\text{sup.th}$ winding turn of the second electrode.

(53) A bending length $fd_{\text{sub.2},k^*+1}$ of the fourth portion at a relative radial position $R_{\text{sub.2},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn of the second electrode may be shorter than a radial length from a relative radial position $R_{\text{sub.2},1}$ of a $1.\text{sup.st}$ winding turn of the second electrode to a relative radial position $R_{\text{sub.1},k^*}$ of a $k^{\text{sup.th}}$ winding turn of the second electrode.

(54) A radius of the core of the electrode assembly may be defined as $r_{\text{sub.c}}$, and a region from a center of the core to $0.90r_{\text{sub.c}}$ may not be blocked by a bent portion of the fourth portion of the second electrode located in a region from a relative radial position $R_{\text{sub.2},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn of the second electrode to a relative radial position of a $n.\text{sub.2}.\text{sup.th}$ winding turn of the first electrode.

(55) A bending length $fd_{\text{sub.2},k^*+1}$ of the fourth portion at the relative radial position $R_{\text{sub.2},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn, the radius $r_{\text{sub.c}}$ of the core, and a distance $d_{\text{sub.2},k^*+1}$ from a center of the electrode assembly to the relative radial position $R_{\text{sub.2},k^*+1}$ may satisfy the following formula:

$$fd_{\text{sub.2},k^*+1} + 0.90 * r_{\text{sub.c}} \leq d_{\text{sub.2},k^*+1}.$$

(56) A region of the second portion from a relative radial position $R_{\text{sub.1},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn of the first electrode to a relative radial position $R_{\text{sub.1},k@}$ of a $k@.\text{sup.th}$ winding turn of the first electrode may be divided into a plurality of segments, and heights of the plurality of segments may increase gradually or stepwise along a direction parallel to a winding direction of the first electrode.

(57) A radial length of the region from the relative radial position $R_{\text{sub.1},k^*+1}$ to the relative radial position $R_{\text{sub.1},k@}$ may be 1% to 56% compared to a radial length of a region from a relative radial position $R_{\text{sub.1},1}$ of a $1.\text{sup.st}$ winding turn of the first electrode to a relative radial position of a $n.\text{sub.1}.\text{sup.th}$ turn of the first electrode.

(58) A region of the second portion from a relative radial position $R_{\text{sub.1},k@+1}$ of a $k@+1.\text{sup.th}$ winding turn of the first electrode to a relative radial position of a $n.\text{sub.1}.\text{sup.th}$ turn of the first electrode may be divided into a plurality of segments, the plurality of segments having substantially a same height from the relative radial position $R_{\text{sub.1},k@+1}$ to the relative radial position of the $n.\text{sub.1}.\text{sup.th}$ turn of the first electrode.

(59) A region of the fourth portion from a relative radial position $R_{\text{sub.2},k^*+1}$ of a $k^*+1.\text{sup.th}$ winding turn of the second electrode to a relative radial position $R_{\text{sub.2},k@}$ of a $k@.\text{sup.th}$ winding turn of the second electrode may be divided into a plurality of segments, and heights of the plurality of segments may increase stepwise or gradually along a direction parallel to a winding direction of the second electrode.

(60) A radial length of the region from the relative radial position $R_{\text{sub.2},k^*+1}$ to the relative radial position $R_{\text{sub.2},k@}$ may be 1% to 56% compared to a radial length of a region from a relative radial position $R_{\text{sub.2},1}$ of a $1.\text{sup.st}$ winding turn of the second electrode to a relative radial position of a $n.\text{sub.2}.\text{sup.th}$ turn of the second electrode.

(61) A region of the fourth portion from a relative radial position $R_{\text{sub.2},k@+1}$ of a $k@+1.\text{sup.th}$ winding turn of the second electrode to a relative radial position of a $n.\text{sub.2}.\text{sup.th}$ turn of the second electrode may be divided into a plurality of segments, and the plurality of segments may

have substantially a same height from the relative radial position R.sub.2,k@+1 to the relative radial position of the n.sub.2.sup.th turn of the second electrode.

(62) A region of the second portion may be divided into a plurality of segments that are independently bendable, and one or more heights of the plurality of segments in the first axial direction or one or more widths of the plurality of segments in a winding direction of the first electrode may increase gradually or stepwise along a direction parallel to the winding direction of the first electrode individually or in groups.

(63) A region of the fourth portion is divided into a plurality of segments that are independently bendable, and one or more heights of the plurality of segments in the second axial direction or one or more widths of the plurality of segments in a winding direction of the second electrode may increase gradually or stepwise along a direction parallel to the winding direction of the second electrode individually or in groups.

(64) Each of the plurality of segments of the second portion or the fourth portion may have a width condition of 1 to 11 mm in the radial direction, a height condition of 2 to 10 mm in the first axial direction or the second axial direction, or a separation pitch condition of 0.05 mm to 1 mm in the winding direction.

(65) The electrode assembly may further include a cut groove between the plurality of segments of the second portion or the fourth portion; and a predetermined gap between a bottom of the cut groove.

(66) The predetermined gap may have a length of 0.2 mm to 4 mm.

(67) The plurality of segments of the second portion or the fourth portion may include a plurality of segment groups along a winding direction of the first electrode or the second electrode, and segments belonging to a same segment group of the plurality of segment groups may have a substantially the same width in the winding direction, height in the first axial direction or the second axial direction, or separation pitch in the winding direction.

(68) The segments belonging to the same segment group may be formed with the width in the winding direction, the height in the first axial direction or the second axial direction, or the separation pitch in the winding direction increasing gradually or stepwise along a direction parallel to the winding direction of the first electrode or the second electrode.

(69) At least a part of the plurality of segment groups may be disposed at a same winding turn of the electrode assembly.

(70) The first surface region may include a stack number increasing region and a stack number uniform region from the outer circumference of the electrode assembly to the core, the stack number increasing region may be a region having an increasing number of the stacked layers of the second portion toward to the core of the electrode assembly, the stack number uniform region may be a region from a radial position where the increasing number of the stacked layers of the second portion stops to a radial position where the second portion starts to bend, and a radial length of the stack number uniform region may be 30% or more compared to a radial length from a winding turn where the second portion starts bending to a winding turn where the second portion stops bending.

(71) The second surface region may include a stack number increasing region and a stack number uniform region from the outer circumference of the electrode assembly to the core, the stack number increasing region may be a region having an increasing number of stacked layers of the fourth portion toward to the core of the electrode assembly, the stack number uniform region may be a region from a radial position where the increasing number of the stacked layers of the fourth portion stops to a radial position where a number of stacked layers of the fourth portion starts to bend, and a radial length of the stack number uniform region may be 30% or more compared to a radial length from a winding turn where the fourth portion starts bending to a winding turn where the fourth portion stops bending.

(72) The first electrode and the second electrode may have a thickness of 80 μm to 250 μm , and an interval of the second portion and the fourth portion at adjacent winding turns in the radial direction

of the electrode assembly may be 200 μm to 500 μm .

(73) The second portion of the first electrode may have a thickness of 10 μm to 25 μm .

(74) The fourth portion of the second electrode may have a thickness of 5 μm to 20 μm .

(75) In the partial region of the first surface region, a total stack thickness of the stacked layers of the second portion may be 100 μm to 975 μm .

(76) The second portion may include a plurality of segments that are independently bendable, the first electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segments in the height uniform region may be 1.0% to 16.3%.

(77) In the partial region of the second surface region, a total stack thickness of stacked layers of the fourth portion may be 50 μm to 780 μm .

(78) The fourth portion may include a plurality of segments that are independently bendable, the second electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and a ratio of a stack thickness of the fourth portion of the second surface region to the substantially uniform height of the plurality of segments in the height uniform region may be 0.5% to 13.0%.

(79) A ratio between a length of a radial region having 10 or more stacked layers of the second portion and a length of the first surface region may be 30% or more.

(80) In another aspect of the present disclosure, there is also provide an electrode assembly that includes a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region, and wherein, in a partial region of the first surface region, a stack thickness of the second portion may be 100 μm to 975 μm .

(81) The first portion of the first electrode may include a plurality of segments that are independently bendable, the first electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region with the plurality of segments having a substantially uniform height, and a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segments in the height uniform region may be 1.0% to 16.3%.

(82) The second electrode may include a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, a part of the fourth portion is bent in the radial direction of the electrode assembly forming a second surface region, and wherein, in a partial region of the second surface region, a stack thickness of the fourth portion may be 50 μm to 780 μm .

(83) The fourth portion may include a plurality of segments that are independently bendable, the second electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and a ratio of a stack thickness of the fourth portion of the second surface region to the substantially uniform height of the segment may be 0.5% to 13.0%

(84) In another aspect of the present disclosure, there is also provided a battery including an electrode assembly having a first electrode, a second electrode, and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator

wound about an axis defining a core and an outer circumference, wherein at least one of the first electrode or the second electrode may include a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, and wherein at least a part of the second portion may be bent in a radial direction of the electrode assembly forming a first surface region including stacked layers of the second portion, and wherein, in a partial region of the first surface region, a number of the stacked layers of the second portion may be 10 or more; a battery housing may accommodate the electrode assembly and may be electrically connected to one of the first electrode or the second electrode to have a first polarity; a sealing body may seal a first opening of the battery housing; a terminal may be electrically connected to the other one of the first electrode or the second electrode to have a second polarity, the terminal having a surface exposed to an outside of the battery housing; and a current collector may be welded to the first surface region and electrically connected to one of the battery housing or the terminal, and a welding region of the current collector overlaps with the first surface region having 10 or more of the stacked layers of the second portion.

(85) A number of total winding turns of the first electrode may be defined as $n_{\text{sub.1}}$, a relative radial position $R_{\text{sub.1},k}$ may be defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the first electrode by the number of total winding turns $n_{\text{sub.1}}$, k being a natural number of 1 to $n_{\text{sub.1}}$, and a ratio between a length of a radial region including the relative radial position $R_{\text{sub.1},k}$ having 10 or more stacked layers of the second portion and a length of the first surface region may be 30% or more.

(86) The second electrode may include a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion may be exposed beyond the separator along a second axis direction of the electrode assembly, a number of total winding turns of the second electrode may be defined as $n_{\text{sub.2}}$, a relative radial position $R_{\text{sub.2},k}$ may be defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the first electrode by the number of total winding turns $n_{\text{sub.2}}$, k being a natural number of 1 to $n_{\text{sub.2}}$, and a ratio between a length of a radial region including the relative radial position $R_{\text{sub.2},k}$ having 10 or more stacked layers of the fourth portion and a length of the second surface region may be 30% or more. 50% or more of the welding region of the current collector may overlap with the first surface region having 10 or more stacked layers of the second portion.

(87) A welding strength of the current collector may be 2 kgf/cm^{sup.2} or more.

(88) In yet another aspect of the present disclosure, there is also provided a battery including an electrode assembly including a first electrode, a second electrode, and a separator between the first electrode and the second electrode, wherein the first electrode, the second electrode, and the separator are wound about an axis to define a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly to form a first surface region, and wherein, in a partial region of the first surface region, a stack thickness of the second portion is 100 μm to 975 μm ; a battery housing accommodating the electrode assembly and electrically connected to one of the first electrode or the second electrode to have a first polarity; a sealing body sealing a first end of the battery housing; a terminal electrically connected to the other of the first electrode or the second electrode to have a second polarity and configured to have a surface exposed to the outside of the battery housing; and a first current collector welded to the first surface region and electrically connected to one of the battery housing or the terminal, wherein a welding region of the first current collector overlaps with the partial region of the first surface region having the stack thickness of the second portion may be 100 μm to 975 μm .

(89) The second portion of the first electrode may be divided into a plurality of segments that are

independently bendable, the first electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segment in the height uniform may be 1.0% to 16.3%.

(90) A welding strength of the first current collector may be 2 kgf/cm² or more.

(91) The second electrode may include a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, a part of the fourth portion may be bent in the radial direction of the electrode assembly forming a second surface region, in a partial region of the second surface region, a stack thickness of the fourth portion may be 50 μ m to 780 μ m, the battery further includes a second current collector welded to the second surface region and electrically connected to the other of the battery housing or the terminal, and a welding region of the second current collector may overlap with the partial region of the second surface region in which the stack thickness of the fourth portion may be 50 μ m to 780 μ m.

(92) The fourth portion of the second electrode may be divided into a plurality of segments that are independently bendable, the second electrode may include a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and a ratio of a stack thickness of the fourth portion in the second surface region to the substantially uniform height of the plurality of segments in the height uniform region may be 0.5% to 13.0%

(93) A welding strength of the second current collector is 2 kgf/cm² or more.

(94) The welding region of the first current collector may overlap with the partial region of the first surface region in which the stack thickness of the second portion is 100 μ m to 975 μ m by 50% or more.

(95) The welding region of the second current collector may overlap with the partial region of the second bending surface region in which the stack thickness of the fourth portion may be 50 μ m to 780 μ m by 50% or more.

(96) In another aspect of the present disclosure, there is also provided a battery pack comprising the battery described above, and a vehicle comprising the battery pack.

(97) Finally, in another aspect of the present disclosure, a method of making an electrode assembly for a battery includes providing a first electrode; providing a second electrode; providing a separator between the first electrode and the second electrode, winding the first electrode, the second electrode, and the separator about an axis to define a core and an outer circumference of the electrode assembly; coating a first portion on the first electrode with an active material; exposing a second portion on the first electrode beyond the separator along a first axial direction of the electrode assembly; bending a part of the second portion in a radial direction of the electrode assembly to form a first surface region including stacked layers of the second portion; and forming in a partial region of the first surface region, 10 or more stacked layers of the second portion in the first axial direction.

(98) The method further includes coating a third portion on the second electrode with the active material; exposing a fourth portion on the first electrode beyond the separator along a second axial direction of the electrode assembly; bending a part of the fourth portion in a radial direction of the electrode assembly to form a second surface region including stacked layers of the fourth portion; and forming in a partial region of the second surface region, 10 or more stacked layers of the fourth portion in the second axial direction.

Advantageous Effects

(99) According to an embodiment of the present disclosure, when bending the uncoated portions exposed at both ends of the electrode assembly, it is possible to prevent the separator or the active material layer from being damaged even if the welding power is increased by sufficiently securing

an area where the uncoated portion is overlapped into 10 or more layers in the radial direction of the electrode assembly.

(100) According to still another embodiment of the present disclosure, since the structure of the uncoated portion adjacent to the core of the electrode assembly is improved, it is possible to prevent the cavity in the core of the electrode assembly from being blocked when the uncoated portion is bent. Thus, the electrolyte injection process and the process of welding the battery housing and the current collector may be carried out easily.

(101) According to still another embodiment of the present disclosure, since the bending surface region of the uncoated portion is directly welded to the current collector instead of a strip-shaped electrode tab, it is possible to provide an electrode assembly with improved energy density and reduced resistance.

(102) According to still another embodiment of the present disclosure, it is possible to provide a battery having a structure that has a low internal resistance and improves welding strength between the current collector and the uncoated portion, and a battery pack and a vehicle including the battery.

(103) In addition, the present disclosure may have several other effects, and such effects will be described in each embodiment, or any description that can be easily inferred by a person skilled in the art will be omitted for an effect.

Description

DESCRIPTION OF DRAWINGS

(1) The accompanying drawings illustrate embodiments of the present disclosure and together with the foregoing disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus, the present disclosure is not construed as being limited to the drawing.

(2) FIG. 1 is a plan view showing a structure of an electrode used for manufacturing a conventional tab-less cylindrical battery.

(3) FIG. 2 is a diagram showing an electrode winding process of the conventional tab-less cylindrical battery.

(4) FIG. 3 is a diagram showing a process of welding a current collector to a bending surface region of an uncoated portion in the conventional tab-less cylindrical battery.

(5) FIG. 4 is a plan view showing a structure of an electrode according to an embodiment of the present disclosure.

(6) FIG. 5 is a diagram showing the definitions of width, height and separation pitch of a segment according to an embodiment of the present disclosure.

(7) FIG. 6 is a diagram for explaining the overlapping condition of segments according to an embodiment of the present disclosure.

(8) FIGS. 7a and 7b are diagrams showing an upper cross-sectional structure and a lower cross-sectional structure of an electrode assembly before the bent structure of an uncoated portion is formed according to an embodiment of the present disclosure, respectively.

(9) FIGS. 8a and 8b are a sectional view and a perspective view showing an electrode assembly in which the uncoated portion is bent to form a bending surface region according to an embodiment of the present disclosure, respectively.

(10) FIG. 9a is a sectional view showing an electrode assembly with a radius of 22 mm included in a cylindrical battery with a form factor of 4680, in which segments of a first electrode are overlapped in a radial direction to form a bending surface region when the segments are bent from the outer circumference toward the core without being overlapped in a circumferential direction.

(11) FIG. 9b is a sectional view showing an electrode assembly with a radius of 22 mm included in the cylindrical battery with a form factor of 4680, in which the segments of the first electrode are

overlapped in the radial direction and in the circumferential direction to form a bending surface region, when the segments are bent from the outer circumference toward the core while being overlapped in the circumferential direction.

(12) FIG. **10** is a sectional view showing a cylindrical battery according to an embodiment of the present disclosure, taken along the Y-axis direction.

(13) FIG. **11** is a sectional view showing a cylindrical battery according to another embodiment of the present disclosure, taken along the Y-axis direction.

(14) FIG. **12** is a plan view showing a structure of a first current collector according to an embodiment of the present disclosure.

(15) FIG. **13** is a perspective view showing a structure of a second current collector according to an embodiment of the present disclosure.

(16) FIG. **14** is a plan view showing a state in which a plurality of cylindrical batteries are electrically connected according to an embodiment of the present disclosure.

(17) FIG. **15** is a partially enlarged plan view showing the electrical connection of the plurality of cylindrical batteries of FIG. **14** in detail.

(18) FIG. **16** is a diagram schematically showing a battery pack including the cylindrical battery according to an embodiment of the present disclosure.

(19) FIG. **17** is a diagram showing a vehicle including the battery pack according to an embodiment of the present disclosure.

BEST MODE

(20) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms used in the specification and the appended claims should not be construed as limited to general and dictionary meanings, but interpreted based on the meanings and concepts corresponding to technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define terms appropriately for the best explanation.

(21) Therefore, the description provided herein are just examples for the purpose of illustrations only, not intended to limit the scope of the disclosure, so it should be understood that other equivalents and modifications could be made thereto without departing from the scope of the disclosure.

(22) First, an electrode assembly according to an embodiment of the present disclosure will be described. The electrode assembly is a jelly-roll type electrode assembly in which a first electrode and a second electrode having a sheet shape and a separator interposed therebetween are wound based on one axis. However, the present disclosure is not limited to the specific type of the electrode assembly and thus the electrode assembly may have any rolled structure known in the art.

(23) Preferably, at least one of the first electrode and the second electrode includes an uncoated portion not coated with an active material at a long side end in the winding direction. At least a part of the uncoated portion is used as an electrode tab by itself.

(24) FIG. **4** is a plan view showing a structure of an electrode **40** according to an embodiment of the present disclosure.

(25) Referring to FIG. **4**, the electrode **40** includes a current collector **41** made of metal foil and an active material layer **42**. The metal foil may be aluminum or copper, and is appropriately selected according to the polarity of the electrode **40**. The active material layer **42** is formed on at least one surface of the current collector **41**, and includes an uncoated portion **43** at the long side end in the winding direction X. The uncoated portion **43** is an area where the active material is not coated. An insulating coating layer **44** may be formed at a boundary between the active material layer **42** and the uncoated portion **43**. The insulating coating layer **44** is formed such that at least a part thereof overlaps with the boundary between the active material layer **42** and the uncoated portion **43**. The insulating coating layer **44** may include a polymer resin and may include an inorganic filler such as Al.sub.2O.sub.3. The region of the uncoated portion **43** in which the insulating coating layer **44** is

formed also corresponds to the uncoated portion **43** because there is no active material layer **42**.

(26) Preferably, a bending part of the uncoated portion **43** of the electrode **40** may include a plurality of segments **61**. The plurality of segments **61** may have a height increasing stepwise from the core toward the outer circumference. The region in which the height increases stepwise is the remaining region except for the uncoated portion adjacent to the core of the electrode assembly (a core-side uncoated portion A). Preferably, the core-side uncoated portion A has a relatively lower height than the other portions.

(27) The segment **61** may be formed by laser notching. The segment **61** may be formed by a known metal foil cutting process such as ultrasonic cutting or punching.

(28) When the electrode **40** is wound, each segment **61** may be bent in the radial direction of the electrode assembly, for example, toward the core, at a bending line **62**. The core refers to the cavity at the winding center of the electrode assembly. Each segment **61** has a geometric shape using the bending line **62** as a base. In the geometric shape, the width of a lower portion thereof may be greater than the width of an upper portion thereof. Also, in the geometric shape, the width of the lower portion may increase gradually or stepwise toward the upper portion. Preferably, the geometric shape may have a trapezoidal shape.

(29) In a modified example, the geometric shape may be formed by connecting one or more straight lines, one or more curves, or a combination thereof. In one example, the geometric shape may be a polygon such as a triangle, a rectangle, or a parallelogram. In another example, the geometric shape may have an arc shape such as a semicircle, semi-ellipse, or the like.

(30) In order to prevent the active material layer **42** and/or the insulating coating layer **44** from being damaged during bending of the segment **61**, it may be preferable to provide a predetermined gap between the bottom (a portion indicated by D4 in FIG. 5) of the cut groove between the segments **61** and the active material layer **42**. This is because stress is concentrated near the bottom of the cut groove when the uncoated portion **43** is bent. The gap may be 0.2 mm to 4 mm. If the gap is adjusted within the corresponding numerical range, it is possible to prevent the active material layer **42** and/or the insulating coating layer **44** from being damaged near the bottom of the cut groove by the stress generated during bending of the segment **61**. In addition, the gap prevents the active material layer **42** and/or the insulating coating layer **44** from being damaged due to tolerances during notching or cutting of the segments **61**.

(31) The plurality of segments **61** may form a plurality of segment groups from the core to the outer circumference. The width, height and separation pitch of segments belonging to the same segment group may be substantially the same.

(32) FIG. 5 is a diagram showing the definitions of width, height and separation pitch of the segment **61** according to an embodiment of the present disclosure.

(33) Referring to FIG. 5, a cut groove **63** is formed between the segments **61**. An edge of the lower portion of the cut groove **63** has a round shape. That is, the cut groove **63** includes a substantially flat bottom portion **63a** and a round portion **63c**. The round portion **63c** connects the bottom portion **63a** and the side **63b** of the segment **61**. In a modified example, the bottom portion **63a** of the cut groove **63** may be replaced with an arc shape. In this case, the sides **63b** of the segments **61** may be smoothly connected by the arc shape of the bottom portion **63a**.

(34) The curvature radius of the round portion **63c** may be greater than 0 and less than or equal to 0.5 mm, more particularly, greater than 0 and less than or equal to 0.1 mm. The round portion **63c** may have a curvature radius of 0.01 mm to 0.05 mm. When the curvature radius of the round portion **63c** meets the above numerical range, it is possible to prevent cracks from occurring in the lower portion of the cut groove **63** while the electrode **40** is traveling in the winding process or the like.

(35) The width (D1), height (D2) and separation pitch (D3) of the segment **61** are designed to prevent abnormal deformation of the uncoated portion **43** as much as possible while sufficiently increasing the number of stacked layers of the uncoated portion **43** in order to prevent the uncoated

portion **43** from being torn during bending of the uncoated portion **43** and improve welding strength of the uncoated portion **43**. Abnormal deformation means that the uncoated portion below the bending point does not maintain a straight state but sinks down to be deformed irregularly. The bending point may be a point spaced apart by 2 mm or less, preferably 1 mm or less, from the bottom of the cut groove **63** indicated by **D4**.

(36) The width (**D1**) of the segment **61** is defined as a length between two points where two straight lines extending from both sides **63b** of the segment **61** meet a straight line extending from the bottom portion **63a** of the cut groove **63**. The height of the segment **61** is defined as the shortest distance between the uppermost side of the segment **61** and a straight line extending from the bottom portion **63a** of the cut groove **63**. The separation pitch (**D3**) of the segment **61** is defined as a length between two points where a straight line extending from the bottom portion **63a** of the cut groove **63** meets straight lines extending from two sidewalls **63b** connected to the bottom portion **63a**. When the side **63b** and/or the bottom portion **63a** is curved, the straight line may be replaced with a tangent line extending from the side **63b** and/or the bottom portion **63a**.

(37) Preferably, the width (**D1**) of the segment **61** may be adjusted in the range of 1 mm to 11 mm. If **D1** is less than 1 mm, a non-overlapping area or an empty space (gap) is generated, thereby not to sufficiently secure welding strength when the segment **61** is bent toward the core. Meanwhile, if **D1** exceeds 11 mm, there is a possibility that the uncoated portion **43** near the bending point (**D4**) is torn due to stress when the segment **61** is bent. The bending point **D4** may be spaced apart from the bottom portion **63a** of the cut groove **63**. The separation distance may be 2 mm or less, preferably 1 mm or less. Also, the height of the segment **61** may be adjusted in the range of 2 mm to 10 mm. If **D2** is less than 2 mm, a non-overlapping area or an empty space (gap) may be generated, thereby not to sufficiently secure welding strength when the segment **61** is bent toward the core.

Meanwhile, if **D2** exceeds 10 mm, it is difficult to manufacture an electrode while uniformly maintaining the flatness of the uncoated portion in the winding direction X. That is, the over-height of the uncoated portion causes cambered surface in the uncoated portion. In addition, the separation pitch (**D3**) of the segment **61** may be adjusted in the range of 0.05 mm to 1 mm. If **D3** is less than 0.05 mm, a crack may occur at the uncoated portion **43** near the bottom of the cut groove **63** due to stress when the electrode **40** travels in the winding process or the like. Meanwhile, if **D3** exceeds 1 mm, a non-overlapping area where the segments **61** do not overlap each other or an empty space (gap) may be generated, thereby not to sufficiently secure welding strength when the segment **61** is bent.

(38) Meanwhile, when the current collector **41** of the electrode **40** is made of aluminum, it is preferable to set the separation pitch **D3** as 0.5 mm or more. When **D3** is 0.5 mm or more, cracks may be prevented from occurring in the lower portion of the cut groove **63** even if the electrode **40** travels at a speed of 100 mm/sec or more under a tension of 300 gf or more in the winding process or the like.

(39) According to the experimental results, when the current collector **41** of the electrode **40** is an aluminum foil with a thickness of 15 μm and **D3** is 0.5 mm or more, cracks do not occur in the lower portion of the cut groove **63** when the electrode **40** travels under the above traveling conditions.

(40) Referring to FIG. 4 again, the width (d.sub.A) of the core-side uncoated portion A is designed by applying a condition that it does not cover the core of the electrode assembly by 90% or more when the segments **61** are bent toward the core.

(41) In one example, the width (d.sub.A) of the core-side uncoated portion A may increase in proportion to the bending length of the segment **61** of Group 1. The bending length corresponds to the height of the segment **61** based on the bending point **62** (FIG. 4).

(42) In a specific example, when the electrode **40** is used to manufacture an electrode assembly of a cylindrical battery having a form factor of 4680, the width (d.sub.A) of the core-side uncoated portion A may be set to 180 mm to 350 mm according to the diameter of the core of the electrode

assembly.

(43) The ratio $d_{\text{sub.A}}/L_{\text{sub.e}}$ of the width ($d_{\text{sub.A}}$) of the core-side uncoated portion (A) to the long side length ($L_{\text{sub.e}}$) of the electrode **40** may be 1% to 30%. In a large-size cylindrical battery with a diameter of 46 mm, the length of the electrode **40** is quite long from 3000 mm to 5000 mm, so the core-side uncoated portion (A) may be designed long enough. In cylindrical batteries with a form factor of 1865 or 2170, the electrode length is in the range of 600 mm to 1200 mm. In a typical cylindrical battery, it is difficult to design the ratio $d_{\text{sub.A}}/L_{\text{sub.e}}$ within the above numerical range.

(44) In an embodiment, the width of each segment group may be designed to constitute the same winding turn of the electrode assembly.

(45) In another embodiment, the width of each segment group may be designed to constitute a plurality of winding turns of the electrode assembly.

(46) In one modification, the width and/or height and/or separation pitch of the segment **61** belonging to the same segment group may be increased or decreased gradually and/or stepwise and/or irregularly within the group or between the groups.

(47) Groups 1 to 7 are only an example of segment groups. The number of groups and the number of segments **61** included in each group may be adjusted to disperse stress as much as possible during the bending process of the uncoated portion **43**, to sufficiently secure the welding strength, to minimize the gap between the sides **63b** of the segments **61**, and to allow the segments **61** to be overlapped into multiple layers along the radial direction of the electrode assembly without interfering with each other.

(48) In one modification, segments of some groups may be removed. In this case, the uncoated portion in a region from which the segments are removed may be the same height as the core-side uncoated portion A.

(49) The electrode **40** may be divided into a height variable region in which the height of the segment **61** changes along the long side direction and a height uniform region in which the height of the segment **61** is uniform.

(50) In the electrode **40**, the height variable region is a region corresponding to Groups 1 to 7, and the height uniform region is a region located near the outer circumference rather than Group 7.

(51) In a specific example, the width ($d_{\text{sub.A}}$) of the core-side uncoated portion A may be 180 mm to 350 mm. The width of Group 1 may be 35% to 55% of the width of the core-side uncoated portion A. The width of Group 2 may be 120% to 150% of the width of Group 1. The width of Group 3 may be 110% to 135% of the width of Group 2. The width of group 4 may be 75% to 90% of the width of group 3. The width of Group 5 may be 120% to 150% of the width of Group 4. The width of Group 6 may be 100% to 120% of the width of Group 5. The width of Group 7 may be 90% to 120% of the width of Group 6.

(52) The reason that the widths of Groups 1 to 7 do not show a constant increase or decrease pattern is that the segment width gradually increases from Group 1 to Group 7, but the number of segments included in the group is limited to an integer number and the thickness of the electrode **40** has a deviation along a winding direction X. Accordingly, the number of segments may be reduced in a specific segment group. Therefore, the widths of the groups may show an irregular change pattern as in the above example from the core to the outer circumference.

(53) Assuming that the width in the winding direction for each of the three segment groups consecutively adjacent to each other in the circumferential direction of the electrode assembly is W_1 , W_2 , and W_3 , respectively, it is possible to include a combination of segment groups in which W_3/W_2 is smaller than W_2/W_1 .

(54) In the specific example, Groups 4 to 6 corresponds to this. The width ratio of Group 5 to Group 4 is 120% to 150%, and the width ratio of Group 6 to Group 5 is 100% to 120%, which is smaller than 120% to 150%.

(55) Preferably, in the plurality of segments **61**, the lower internal angle (θ) may increase from the

core to the outer circumference. The lower internal angle (θ) corresponds to an angle between the straight line passing through the bending line **62** (FIG. **4**) and the straight line (or, the tangent line) extending from the side **63b** of the segment **61**. When the segment **61** is asymmetrical in the left and right direction, the left internal angle and the right internal angle may be different from each other.

(56) As the radius of the electrode assembly increases, the radius of curvature increases. If the lower internal angle (θ) of the segment **61** increases as the radius of the electrode assembly increases, the stress generated in the radial and circumferential directions when the segment **61** is bent may be relieved. In addition, when the lower internal angle (θ) is increased, when the segment **61** is bent, the area overlapping with the segment **61** at an inner side and the number of stacked layers of the segment **61** also increase, thereby securing uniform welding strength in the radial and circumferential directions and making the bending surface region flat.

(57) Preferably, if the angle of the lower internal angle (θ) is adjusted as the radius of the electrode assembly increases, when the segments **61** are bent, the segments **61** may be overlapped in the circumferential direction as well as in the radial direction of the electrode assembly.

(58) FIG. **6** shows an example in which the sides of the segments **61** bent toward the core of the electrode assembly are spaced apart in parallel in an arbitrary winding turn with a radius of r based on the core center and an example in which the sides of the bent segments **61** intersect each other.

(59) Referring to FIG. **6**, a pair of segments **61** adjacent to each other are disposed in a winding turn having a radius of r with respect to the core center O of the electrode assembly. The width and height of the adjacent segments **61** are substantially the same.

(60) In (a) of FIG. **6**, the lower internal angle $\theta_{\text{sub.assumption}}$ is an angle in the assumption that the sides of the segment **61** are substantially parallel. The lower internal angle $\theta_{\text{sub.assumption}}$ is an angle that can be uniquely determined by the arc length $L_{\text{sub.arc}}$ corresponding to the lower portion of the segment **61**. Meanwhile, $\theta_{\text{sub.real}}$ is an actual lower internal angle when the sides of the adjacent segments **61** intersect each other.

(61) Preferably, when the lower internal angles $\theta_{\text{sub.assumption}}$ and $\theta_{\text{sub.real}}$ satisfy Formula 1 below, the segments **61** disposed in the winding turn located at the radius of r with respect to the core center O may overlap with each other in the circumferential direction.

$\theta_{\text{sub.real}} > \theta_{\text{sub.assumption}}$

$\theta_{\text{sub.assumption}} = 90^\circ - 360^\circ * (L_{\text{sub.arc}} / 2\pi r) * 0.5$

$\theta_{\text{sub.real}} > 90^\circ - 360^\circ * (L_{\text{sub.arc}} / 2\pi r) * 0.5$ <Formula 1>

(62) Here, r is a radius of the winding turn where the segment **61** is disposed based on the core center of the electrode assembly.

(63) $L_{\text{sub.arc}}$ is a length of the arc (solid line) corresponding to the lower portion (dotted line) of the segment in a circle with a radius of r , and is uniquely determined from the width ($D1$) of the segment **61**.

(64) ' $360^\circ * (L_{\text{sub.arc}} / 2\pi r)$ ' is a circumferential angle α of the lower portion (dotted line) of the segment **61**.

(65) ' $360^\circ * (L_{\text{sub.arc}} / 2\pi r) * 0.5$ ' is an angle between the line segment OB and the line segment OA in the right triangle OAB .

(66) ' $90^\circ - 360^\circ * (L_{\text{sub.arc}} / 2\pi r) * 0.5$ ' is an angle between the line segment OA and the line segment AB in the right triangle OAB , which approximately corresponds to the lower internal angle ($\theta_{\text{sub.assumption}}$) of the segment **61**.

(67) Preferably, the circumferential angle α of $L_{\text{sub.arc}}$ at any winding turn radius r may be less than or equal to 45° . If the circumferential angle α exceeds 45° , the segment **61** is not bent easily. Therefore, at any radius r , $L_{\text{sub.arc}}$ is greater than 1 mm, which is the lower limit of $D1$, and has a length of $(45/360) * (2\pi r)$ or less.

(68) The circumferential angle α may vary depending on a radius (r) of a winding turn at which the segment **61** is located. In one aspect, the circumferential angle α of the segment **61** may increase

gradually or stepwise along a radial direction of the electrode assembly within the above numerical range, or vice versa. In other aspect, the circumferential angle α of the segment **61** may increase gradually or stepwise and then decreases gradually or stepwise along a radial direction of the electrode assembly within the above numerical range, or vice versa. In another aspect, the circumferential angle α of the segment **61** may be substantially the same along a radial direction of the electrode assembly within the above numerical range.

(69) Preferably, when a width of each of the plurality of segments **61** varies along a winding direction, the circumferential angle α may be in the range of 45 degrees or less and the width of each of the plurality of segments **61** may be in the range of 1 mm to 11 mm.

(70) In one example, when r is 20 mm and the circumferential angle α is 30° , $L_{\text{sub.arc}}$ is 10.5 mm and $\theta_{\text{sub.assumption}}$ is about 75 degrees. As another example, if r is 25 mm and the circumferential angle α is 25° , $L_{\text{sub.arc}}$ is 10.9 mm and $\theta_{\text{sub.assumption}}$ is about 77.5 degrees.

(71) Preferably, at any winding turn radius r , $(\theta_{\text{sub.real}}/\theta_{\text{sub.assumption}}-1)$ may be defined as an overlapping ratio of the segment **61** in the circumferential direction. The overlapping ratio of the segment **61** may be greater than 0 and equal to or less than 0.05. $\theta_{\text{sub.assumption}}$ is the angle uniquely determined by the arc $L_{\text{sub.arc}}$ at the winding turn radius r . If the overlapping ratio of the segment **61** is greater than 0.05, when the segments **61** are bent, the sides of the segments **61** may interfere with each other, and thus the segments **61** may not be bent easily.

(72) The degree of overlapping of the segments **61** increases in proportion to the overlapping ratio. If the segments **61** overlap with each other along the circumferential direction of the winding turn, the number of stacked layers of the segments **61** may be further increased when the segments **61** are bent. Embodiments for this will be described later.

(73) Preferably, when the electrode **40** is used to manufacture an electrode assembly of a cylindrical battery having a form factor of 4680, the radius of the core is 4 mm, and the height of the segment closest to the core is 3 mm, when the radius of the electrode assembly increases from 7 mm to 22 mm, the lower internal angle of the segments **61** may increase stepwise in the range of 60° to 85° .

(74) The radius range and the lower internal angle range may be determined from the form factor and the design specifications about the diameter of the core, the height of the segment closest to the core, the width ($D1$) of the segment **61**, and the overlapping ratio.

(75) Meanwhile, the overlapping condition of the segments may be changed as follows. That is, when a virtual circle passing through a pair of segments **61** adjacent to each other based on the core center O of the electrode assembly **40** is drawn as shown in (b) of FIG., if an arc $e_{\text{sub.1-e.sub.2}}$ and an arc $e_{\text{sub.3-e.sub.4}}$ passing through each segment overlap with each other **61**, the pair of adjacent segments may overlap with each other. The overlapping ratio of the segment **61** may be defined as a maximum value for a ratio of the length of the overlapping arc $e_{\text{sub.2-e.sub.3}}$ to the length of the arc $e_{\text{sub.1-e.sub.2}}$ (or, $e_{\text{sub.3-e.sub.4}}$) when a plurality of virtual circles with different radii are drawn. The overlapping ratio of the segment **61** may be greater than 0 and equal to or less than 0.05.

(76) The shapes of the segments **60** may be changed differently depending locations. In one example, a round shape (e.g., semicircle, semi-ellipse, etc.) that is advantageous for stress distribution is applied to a region where the stress is concentrated, and a polygonal shape (e.g., a rectangle, trapezoid, parallelogram, etc.) with the largest area may be applied a region where the stress is relatively low.

(77) The segment structure may also be applied to the core-side uncoated portion A. However, if the segment structure is applied to the core-side uncoated portion A, when the segments are bent according to the radius of curvature of the core, the end of the core-side uncoated portion A may be bent toward the outer circumference, which is called reverse forming. Therefore, the core-side uncoated portion A has no segment, or even if the segment structure is applied to the core-side uncoated portion A, it is desirable to control the width and/or height and/or separation pitch of the

segments 61 in consideration of the radius of curvature of the core such that reverse forming does not occur.

(78) The electrode structure of the above embodiments (modifications) may be applied to the first electrode and/or the second electrode having different polarities included in the jelly-roll type electrode assembly. In addition, when the electrode structure of the above embodiments (modifications) is applied to any one of the first electrode and the second electrode, the conventional electrode structure may be applied to the other one. In addition, the electrode structures applied to the first electrode and the second electrode may not be identical but be different from each other.

(79) For example, when the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, any one of the above embodiments (modifications) may be applied to the first electrode and the conventional electrode structure (see FIG. 1) may be applied to the second electrode.

(80) As another example, when the first electrode and the second electrode are a positive electrode and a negative electrode, respectively, any one of the above embodiments (modifications) may be selectively applied to the first electrode and any one of the above embodiments (modifications) may be selectively applied to the second electrode.

(81) In the present disclosure, a positive electrode active material coated on the positive electrode and a negative electrode active material coated on the negative electrode may employ any active material known in the art without limitation.

(82) In one example, the positive electrode active material may include an alkali metal compound expressed by a general formula $A[A_{x_1}M_{y_1}O]_{2+z}$ (A includes at least one element among Li, Na and K; M includes at least one element selected from is Ni, Co, Mn, Ca, Mg, Al, Ti, Si, Fe, Mo, V, Zr, Zn, Cu, Al, Mo, Sc, Zr, Ru, and Cr; $x \geq 0$, $1 \leq x+y \leq 2$, $-0.1 \leq z \leq 2$; and the stoichiometric coefficients x, y and z are selected so that the compound maintains electrical neutrality).

(83) In another example, the positive electrode active material may be an alkali metal compound $xLiM_{sup.1}O_{sub.2}-(1-x)Li_{sub.2}M_{sup.2}O_{sub.3}$ disclosed in U.S. Pat. Nos. 6,677,082, 6,680,143, et al., wherein $M_{sup.1}$ includes at least one element having an average oxidation state 3; $M_{sup.2}$ includes at least one element having an average oxidation state 4; and $0 \leq x \leq 1$).

(84) In still another example, the positive electrode active material may be lithium metal phosphate expressed by a general formula $Li_{sub.a}M_{sup.1sub.x}Fe_{sub.1-x}M_{sup.2sub.y}P_{sub.1-y}M_{sup.3sub.z}O_{sub.4-z}$ ($M_{sup.1}$ includes at least one element selected from the Ti, Si, Mn, Co, Fe, V, Cr, Mo, Ni, Nd, Al, Mg and Al; $M_{sup.2}$ includes at least one element selected from Ti, Si, Mn, Co, Fe, V, Cr, Mo, Ni, Nd, Al, Mg, Al, As, Sb, Si, Ge, V and S; $M_{sup.3}$ includes a halogen element optionally including F; $0 < a \leq 2$, $0 \leq x \leq 1$, $0 \leq y < 1$, $0 \leq z < 1$; the stoichiometric coefficient a, x, y and z are selected so that the compound maintains electrical neutrality), or $Li_{sub.3}M_{sub.2}(PO_{sub.4})_{sub.3}$ (M includes at least one element selected from Ti, Si, Mn, Fe, Co, V, Cr, Mo, Ni, Al, Mg and Al).

(85) Preferably, the positive electrode active material may include primary particles and/or secondary particles in which the primary particles are aggregated.

(86) In one example, the negative electrode active material may employ carbon material, lithium metal or lithium metal compound, silicon or silicon compound, tin or tin compound, or the like. Metal oxides such as $TiO_{sub.2}$ and $SnO_{sub.2}$ with a potential of less than 2V may also be used as the negative electrode active material. As the carbon material, low-crystalline carbon, high-crystalline carbon or the like may be used.

(87) The separator may employ a porous polymer film, for example, a porous polymer film made of a polyolefin-based polymer such as ethylene homopolymer, propylene homopolymer, ethylene/butene copolymer, ethylene/hexene copolymer, ethylene/methacrylate copolymer, or the like, or laminates thereof. As another example, the separator may employ a common porous

nonwoven fabric, for example, a nonwoven fabric made of high melting point glass fiber, polyethylene terephthalate fiber, or the like.

(88) A coating layer of inorganic particles may be included in at least one surface of the separator. It is also possible that the separator itself is made of a coating layer of inorganic particles. Particles in the coating layer may be coupled with a binder so that an interstitial volume exists between adjacent particles.

(89) The inorganic particles may be made of an inorganic material having a dielectric constant of 5 or more. As a non-limiting example, the inorganic particles may include at least one material selected from the group consisting of $\text{Pb}(\text{Zr,Ti})\text{O}_{0.3}$ (PZT), $\text{Pb}_{0.1-x}\text{La}_x\text{Zr}_{0.1-y}\text{Ti}_y\text{O}_{0.3}$ (PLZT), $\text{PB}(\text{Mg}_{0.3}\text{Nb}_{0.2/3})\text{O}_{0.3}$ — $\text{PbTiO}_{0.3}$ (PMN-PT), $\text{BaTiO}_{0.3}$, hafnia ($\text{HfO}_{0.2}$), $\text{SrTiO}_{0.3}$, $\text{TiO}_{0.2}$, $\text{Al}_{0.2}\text{O}_{0.3}$, $\text{ZrO}_{0.2}$, $\text{SnO}_{0.2}$, $\text{CeO}_{0.2}$, MgO , CaO , ZnO and $\text{Y}_{0.2}\text{O}_{0.3}$.

(90) The electrode assembly according to the embodiment is a jelly-roll type electrode assembly **80** in which the electrode **40** of the embodiment is applied to a first electrode (positive electrode) and a second electrode (negative electrode). However, the present disclosure is not limited to specific kind of the electrode assembly.

(91) FIGS. **7a** and **7b** are drawings showing an upper cross-sectional structure and a lower cross-sectional structure of the electrode assembly **80** before the bending structures of the uncoated portions **43a**, **43a'** are formed according to an embodiment of the present disclosure, respectively. In addition, FIGS. **8a** and **8b** are a cross-sectional view and a perspective view showing the electrode assembly **80** in which the bending surface region F is formed while the uncoated portions **43a**, **43a'** are bent according to an embodiment of the present disclosure, respectively.

(92) The electrode assembly **80** may be manufactured by the winding method described with reference to FIG. **2**. For convenience of description, the protruding structures of the uncoated portions **43a**, **43a'** extending out of the separator are illustrated in detail, and the winding structure of the separator is not depicted. The uncoated portion **43a** of the electrode assembly **80** protruding upward extends from the first electrode **40**. The uncoated portion **43a'** of the electrode assembly **80** protruding downward extends from the second electrode **40'**. The end of the separator is marked by a dotted line.

(93) The patterns in which the heights of the uncoated portions **43a**, **43a'** change are schematically illustrated. That is, the heights of the uncoated portions **43a**, **43a'** may vary irregularly depending on the position at which the cross-section is cut. For example, when the sides of the segments **61** having a trapezoidal shape are cut, the height of the uncoated portion in the cross section is lower than the height (D2 in FIG. **4**) of the segments **61**. In addition, the uncoated portions **43a**, **43a'** are not shown at the point where the cut groove **63** (FIG. **5**) is cut.

(94) Hereinafter, the structural features of the uncoated portion **43a** of the first electrode **40** will be described in detail with reference to the drawings. Preferably, the uncoated portion **43a'** of the second electrode **40'** may also have substantially the same characteristics as the uncoated portion **43a** of the first electrode **40**.

(95) Referring to FIGS. **7a**, **7b**, **8a** and **8b**, the uncoated portions **43a**, **43a'** of the first electrode **40** and the second electrode **40'** are bent in the radial direction to form a bending surface region F.

(96) In the winding structure of the first electrode **40**, assuming that the number of total winding turns of the first electrode **40** is $n_{0.1}$, when a value obtained by dividing a winding turn index k (a natural number of 1 to $n_{0.1}$) of a k_{th} winding turn by the number of total winding turns $n_{0.1}$ is defined as a relative radial position $R_{0.1,k}$ of the k_{th} winding turn, a radial length of the relative radial position $R_{0.1,k}$ region (e.g., a region from $R_{0.1,k^*+1}$ to $R_{0.1,K}$) where the number of stacked layers of the uncoated portion **43a** is 10 or more is 30% or more compared to the radial length (e.g., $R_{0.1}$) of winding turns including the segments.

(97) For reference, the relative radial position of the 1st winding turn is $1/n_{0.1}$ because the winding turn index is 1. The relative radial position of the k_{th} winding turn is $k/n_{0.1}$. The

relative radial position of the last $n_{\text{sub.1}}^{\text{sup.th}}$ winding turn is 1. That is, the relative radial position increases from $1/n_{\text{sub.1}}$ to 1 from the core of the electrode assembly **80** to the outer circumference thereof.

(98) In the winding structure of the second electrode **40'**, assuming that the number of total winding turns of the second electrode **40'** is $n_{\text{sub.2}}$, when a value obtained by dividing a winding turn index k (a natural number of 1 to $n_{\text{sub.2}}$) at a $k^{\text{sup.th}}$ winding turn location by the number of total winding turns $n_{\text{sub.2}}$ is defined as a relative radial position $R_{\text{sub.2},k}$ of the $k^{\text{sup.th}}$ winding turn, a radial length of the relative radial position $R_{\text{sub.2},k}$ region where the number of stacked layers of the uncoated portion is 10 or more is 30% or more compared to the radial length of winding turns in which the segments are disposed.

(99) For reference, the relative radial position of the $1^{\text{sup.st}}$ winding turn is $1/n_{\text{sub.2}}$ because the winding turn index is 1. The relative radial position of the $k^{\text{sup.th}}$ winding turn is $k/n_{\text{sub.2}}$. The relative radial position of the last $n_{\text{sub.2}}^{\text{sup.th}}$ winding turn is 1. That is, the relative radial position increases from $1/n_{\text{sub.2}}$ to 1 from the core of the electrode assembly **80** to the outer circumference thereof.

(100) Preferably, the winding turn indexes k of the first electrode **40** and the second electrode **40'** should be understood as variables to which different values can be assigned.

(101) When the uncoated portions **43a**, **43a'** are bent in the radial direction, the bending surface regions **F** are formed on the upper and lower portions of the electrode assembly **80** as shown in FIGS. **8a** and **8b**.

(102) Referring to FIGS. **8a** and **8b**, a plurality of segments **61** are overlapped into multiple layers along the radial direction while being bent toward the core **C** of the electrode assembly **80**.

(103) The number of stacked layers of the segments **61** may be defined as the number of segments **61** that intersect an imaginary line when the imaginary line is drawn in the winding axis direction (**Y**) at any radial point on the bending surface region **F**.

(104) Preferably, the number of stacked layers of the segments **61** may be 10 or more in a radius region of at least 30% based on the radial length ($R_{\text{sub.1}}$) of the winding turns including the segments **61** in order to sufficiently increase the welding strength between the bending surface region **F** and the current collector and to prevent the separator and the active material layer from being damaged during the welding process.

(105) The current collector may be laser-welded to the bending surface region **F** of the uncoated portion **43a**, **43a'**. Alternatively, other known welding techniques such as resistance welding may be used. When laser welding is applied, it is desirable to increase the laser power to sufficiently secure the welding strength. If the laser power is increased, the laser may penetrate through the overlapping regions of the uncoated portions **43a**, **43a'** into the electrode assembly **80**, which may damage the separator and the active material layer. Therefore, in order to prevent laser penetration, it is preferable to increase the number of stacked layers of the uncoated portions **43a**, **43a'** in the welding region to a certain level or more. In order to increase the number of stacked layers of the uncoated portions **43a**, **43a'**, the height of the segments **61** should be increased. However, if the height of the segments **61** is increased, the uncoated portions **43a**, **43a'** may be cambered during the manufacturing process of the electrode **40**. Therefore, it is desirable to adjust the height of the segments **61** to an appropriate level, such as 2 mm to 10 mm.

(106) If the radius region where the number of stacked layers of the segments **61** is 10 or more is designed to be FIG. **7** compared to $R_{\text{sub.1}}$ in the bending surface region **F** and the region where the segments **61** are overlapped into 10 or more layers is laser-welded to the current collector, even though the power of laser is increased, the overlapping portion of the uncoated portion sufficiently masks the laser to prevent the separator and the active material layer from being damaged by the laser. In addition, since the number of stacked layers of the segments **61** is large in the area where laser is irradiated, welding beads are formed with sufficient volume and thickness. Accordingly, the welding strength may be sufficiently secured and the resistance of the welding interface may also

be lowered.

(107) When welding the current collector, the laser power may be determined by a desired welding strength between the bending surface region F and the current collector. The welding strength increases in proportion to the number of stacked layers of the uncoated portions **43a**, **43a'**. This is because as the number of stacked layers of the uncoated portions **43a**, **43a'** increases, the volume of the weld beads formed by the laser increases.

(108) The welding strength may be 2 kgf/cm² or more, more particularly, 4 kgf/cm² or more. If the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly **80** along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and the volume of the welding beads is sufficient to reduce the resistance of the welding interface. The power of the laser for realizing the above welding strength condition varies depending on the laser equipment, and may be appropriately adjusted in the range of 250 W to 320 W, or in the range of 40% to 100% of the maximum laser power specification.

(109) The welding strength may be defined as a tensile force per unit area (kgf/cm²) of the current collector when the current collector starts to separate from the bending surface region F. Specifically, after the current collector is completely welded, a tensile force is applied to the current collector, but the magnitude of the tensile force is gradually increased. As the tensile force is increased, the uncoated portions **43a**, **43a'** begin to separate from the welding interface. At this time, the value obtained by dividing the tensile force applied to the current collector by the area of the current collector is the welding strength.

(110) Preferably, the first electrode **40** may include a current collector (foil) **41** and an active material coating layer **42** formed on at least one surface of the current collector **41**. Here, the current collector **41** may have a thickness of 10 μm to 25 μm, and an interval between winding turns adjacent in a radial direction of the electrode assembly **80** may be 200 μm to 500 μm. The current collector **41** may be made of aluminum.

(111) The second electrode **40'** may include a current collector (foil) and an active material coating layer formed on at least one surface of the current collector. Here, the current collector may have a thickness of 5 μm to 20 μm, and an interval between winding turns adjacent in the radial direction of the electrode assembly **80** may be 200 μm to 500 μm. The current collector may be made of copper.

(112) Referring to FIGS. **4**, **7a** and **7b**, in the winding structure of the first electrode **40**, the uncoated portion of the region from the relative radial position R.sub.1,1 of the first electrode **40** to a preset first relative radial position R.sub.1,k* may have a smaller height than the uncoated portion of the region from a relative radial position R.sub.1,k*+1 of a k*+1.sup.th winding turn to a relative radial position **1**. The height of the uncoated portion of the region from the relative radial position R.sub.1,1 to the preset first relative radial position R.sub.1,k* corresponds to the height of the uncoated portion of the core-side uncoated portion A (see FIG. **4**).

(113) Preferably, in the winding structure of the first electrode **40**, the uncoated portion of the region from the relative radial position R.sub.1,1 to the first relative radial position R.sub.1,k* may have a smaller height than the bending surface region F formed by overlapping the bent uncoated portions.

(114) Preferably, in the winding structure of the first electrode **40**, the uncoated portion of the region from the relative radial position R.sub.1,1 to the first relative radial position R.sub.1,k* may not be bent toward the core of the electrode assembly **80**.

(115) Similar to the first electrode **40**, in the winding structure of the second electrode **40'**, the uncoated portion of the region from the relative radial position R.sub.2,1 to the preset first relative radial position R.sub.2,k* may have a smaller height than the uncoated portion of the region from the relative radial position R.sub.2,k*+1 of the k*+1.sup.th winding turn to the relative radial position **1**.

(116) In addition, in the region from the relative radial position $R_{\text{sub.2},1}$ to the preset first relative radial position $R_{\text{sub.2},k^*}$, the uncoated portion may have a smaller height than the bending surface region F formed by overlapping the bent uncoated portions.

(117) Preferably, the uncoated portion of the region from the relative radial position $R_{\text{sub.2},1}$ to the first relative radial position $R_{\text{sub.2},k^*}$ may not be bent towards the core of the electrode assembly.

(118) Preferably, in the winding structure of the second electrode **40'**, the uncoated portion of the region from the relative radial position $R_{\text{sub.2},1}$ to the first relative radial position $R_{\text{sub.2},k^*}$ may have a smaller height than the uncoated portion of the region from the relative radial position $R_{\text{sub.2},k^*+1}$ to the relative radial position **1** and may not be bent toward the core.

(119) In the winding structure of the first electrode **40**, the bending length $fd_{\text{sub.1},k^*+1}$ of the uncoated portion of the relative radial position $R_{\text{sub.1},k^*+1}$ may be shorter than the radial length from the relative radial position $R_{\text{sub.1},1}$ to the relative radial position $R_{\text{sub.1},k^*}$. Therefore, the core C of the electrode assembly **80** may not be blocked by the bending part of the uncoated portion **43a** located in the region from the relative radial position $R_{\text{sub.1},k^*+1}$ to the relative radial position **1**.

(120) Alternatively, the core C of the electrode assembly **80** may not be blocked by the bent portion of the uncoated portion **43a** located in the region from the relative radial position $R_{\text{sub.1},k^*+1}$ to the relative radial position **1** by 90% or more based on its radius ($r_{\text{sub.c}}$) thereof. That is, a radial region of the core C corresponding to at least 0 to 0.9 $r_{\text{sub.c}}$ may not be blocked by the bent portion of the uncoated portion **43a**.

(121) Preferably, the bending length $fd_{\text{sub.1},k^*+1}$ of the uncoated portion **43a** located at the relative radial position $R_{\text{sub.1},k^*+1}$, the radius ($r_{\text{sub.c}}$) of the core, and the distance ($d_{\text{sub.1},k^*+1}$) from the center of the core C to the relative radial position $R_{\text{sub.1},k^*+1}$ may satisfy Formula 2 below.

$$fd_{\text{sub.1},k^*+1} + 0.9 * r_{\text{sub.c}} \leq d_{\text{sub.1},k^*+1} \quad \text{<Formula 2>}$$

(122) Preferably, in the winding structure of the second electrode **40'**, the uncoated portion of the region from the relative radial position $R_{\text{sub.2},1}$ to the first relative radial position $R_{\text{sub.2},k^*}$ may have a smaller height than the uncoated portion of the region from the relative radial position $R_{\text{sub.2},k^*+1}$ to the relative radial position **1** and may not be bent toward the core.

(123) In the winding structure of the second electrode **40'**, the bending length $fd_{\text{sub.2},k^*+1}$ of the uncoated portion located at the relative radial position $R_{\text{sub.2},k^*+1}$ may be shorter than the length from the relative radial position $R_{\text{sub.2},1}$ to the first relative radial position $R_{\text{sub.2},k^*}$. Therefore, the core C of the electrode assembly **80** may not be blocked by the bending part of the uncoated portion located in the region from the relative radial position $R_{\text{sub.2},k^*+1}$ to the relative radial position **1**.

(124) Alternatively, the core C of the electrode assembly **80** may not be blocked by the bent portion of the uncoated portion **43a'** located at the relative radial position $R_{\text{sub.2},k^*+1}$ by 90% or more based on its radius ($r_{\text{sub.c}}$).

(125) Preferably, the bending length $fd_{\text{sub.2},k^*+1}$ of the uncoated portion **43a'** located at the relative radial position $R_{\text{sub.2},k^*+1}$, the radius ($r_{\text{sub.c}}$) of the core, and the distance ($d_{\text{sub.2},k^*+1}$) from the center of the core C to the relative radial position $R_{\text{sub.2},k^*+1}$ may satisfy Formula 3 below.

$$fd_{\text{sub.2},k^*+1} + 0.9 * r_{\text{sub.c}} \leq d_{\text{sub.2},k^*+1} \quad \text{<Formula 3>}$$

(126) Preferably, in the winding structure of the first electrode **40**, the uncoated portion of the second electrode **40'** from a second relative radial position $R_{\text{sub.1},k@+1}$ of the preset $k@+1$.sup.th winding turn to the relative radial position **1** is divided into a plurality of segments **61**, and the height of the plurality of segments **61** may be substantially the same from the relative radial position $R_{\text{sub.1},k@+1}$ to the relative radial position **1**.

(127) Meanwhile, in the winding structure of the first electrode **40**, the uncoated portion **43a** of a

region from the relative radial position $R_{\text{sub.1},k^*+1}$ to the second relative radial position $R_{\text{sub.1},k@}$ of a preset $k@$ sup.th winding turn is divided into a plurality of segments **61**, whose heights may increase stepwise or gradually toward the outer circumference. Therefore, the region from the relative radial position $R_{\text{sub.1},k^*+1}$ to the relative radial position $R_{\text{sub.1},k@}$ corresponds to the height variable region.

(128) For example, in the winding structure of the first electrode **40** with a radius of 22 mm, when the radial length of the height variable region of the segment is defined as $H_{\text{sub.1}}$ and the ratio of $H_{\text{sub.1}}$ to the radius ($R-r_{\text{sub.c}}$) of the winding structure of the first electrode **40** except for the core C is defined as a height variable region ratio ($H_{\text{sub.1}}/(R-r_{\text{sub.c}})$), the ratio of the height variable region may be calculated as follows by rounding off to the zero decimal place.

(129) In Example 1, R may be 22 mm, the core radius ($r_{\text{sub.c}}$) may be 5 mm, and $R-r_{\text{sub.c}}$ may be 17 mm. The height of the segment **61** may be changed in 8 steps from 2 mm to 10 mm in the radius region of 7 mm to 15 mm. After radius 15 mm, the height of the segment **61** is maintained at 10 mm. Since $H_{\text{sub.1}}$ is 8 mm, the height variable region ratio may be 47% (8 mm/17 mm).

(130) In Example 2, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 7 steps from 2 mm to 9 mm in the radius region of 7 mm to 14 mm. After radius 14 mm, the height of the segment **61** is maintained at 9 mm. Since $H_{\text{sub.1}}$ is 7 mm, the height variable region ratio may be 41% (7 mm/17 mm).

(131) In Example 3, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 6 steps from 2 mm to 8 mm in the radius region of 7 mm to 13 mm. After radius 13 mm, the height of the segment **61** is maintained at 8 mm. Since $H_{\text{sub.1}}$ is 6 mm, the height variable region ratio may be 35% (6 mm/17 mm).

(132) In Example 4, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 5 steps from 2 mm to 7 mm in the radius region of 7 mm to 12 mm. After radius 12 mm, the height of the segment **61** is maintained at 7 mm. Since $H_{\text{sub.1}}$ is 5 mm, the height variable region ratio may be 29% (5 mm/17 mm).

(133) In Example 5, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 4 steps from 2 mm to 6 mm in the radius region of 7 mm to 11 mm. After radius 11 mm, the height of the segment **61** is maintained at 6 mm. Since $H_{\text{sub.1}}$ is 4 mm, the height variable region ratio may be 24% (4 mm/17 mm).

(134) In Example 6, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 3 steps from 2 mm to 5 mm in the radius region of 7 mm to 10 mm. After radius 10 mm, the height of the segment **61** is maintained at 5 mm. Since $H_{\text{sub.1}}$ is 3 mm, the height variable region ratio may be 18% (3 mm/17 mm).

(135) In Example 7, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed in 2 steps from 2 mm to 4 mm in the radius region of 7 mm to 9 mm. After radius 9 mm, the height of the segment **61** is maintained at 4 mm. Since $H_{\text{sub.1}}$ is 2 mm, the height variable region ratio may be 12% (2 mm/17 mm).

(136) In Example 8, R and $r_{\text{sub.c}}$ are the same as in Example 1. The height of the segment **61** may be changed stepwise in 1 step from 2 mm to 3 mm in the radius region of 7 mm to 8 mm. After radius 8 mm, the height of the segment **61** is maintained at 3 mm. Since $H_{\text{sub.1}}$ is 1 mm, the height variable region ratio may be 6% (1 mm/17 mm).

(137) In summary, when R is 22 mm and $r_{\text{sub.c}}$ is 5 mm, if the height of the segment in the radius region of 7 mm to 15 mm changes in the range of 2 mm to 10 mm in any one of one to eight steps, the height variable region ratio may be 6% to 47%.

(138) The numerical range of the height variable region ratio may be changed according to the size of the radius ($r_{\text{sub.c}}$) of the core C. Since the calculation method is similar to the above, only the results are disclosed.

(139) In an example, when R is 22 mm and $r_{\text{sub.c}}$ is 4 mm, if the height of the segment in the radius region of 6 mm to 14 mm changes stepwise in the range of 2 mm to 10 mm in any one of

one to eight steps, the height variable region ratio may be 6% to 44%.

(140) In another example, when R is 22 mm and $r_{sub.c}$ is 3 mm, if the height of the segment in the radius region of 5 mm to 13 mm changes stepwise in the range of 2 mm to 10 mm in any one of one to eight steps, the height variable region ratio may be 5% to 42%.

(141) In still another example, when R is 22 mm and $r_{sub.c}$ is 2 mm, if the height of the segment in the radius region of 4 mm to 12 mm changes stepwise in the range of 2 mm to 10 mm in any one of one to eight steps, the height variable region ratio may be 5% to 40%.

(142) From the above calculation examples, when the radius ($r_{sub.c}$) of the core C is changed in the range of 2 mm to 5 mm, the height variable region ratio is 5% to 47%. When the radius of the electrode assembly **80** is constant, the lower and upper limits of the height variable region ratio decrease accordingly as the radius ($r_{sub.c}$) of the core C decreases.

(143) Meanwhile, the upper and lower limits of the height variable region ratio may be changed by the height change amount of the segment **61** per 1 mm increase in radius and the number of height changes.

(144) In one example, when the height of the segment **61** changes by 0.2 mm per 1 mm increase in radius, the lower and upper limits of the height variable region ratio are 1% and 9%, respectively.

(145) In another example, when the height of the segment **61** changes by 1.2 mm per 1 mm increase in radius, the lower and upper limits of the height variable region ratio are 6% and 56%, respectively.

(146) From the above examples, the height variable region ratio may be 1% to 56%. If the height variable region ratio of the segment **61** satisfies the above numerical range, the ratio of relative radial positions where the number of stacked layers of the uncoated portion **40** is 10 or more may be at least 30% of the radial length ($R_{sub.1}$) of the winding turns including the segment **61**. As will be described later, this configuration provides useful effects in terms of welding strength and resistance of the current collector.

(147) Referring to FIGS. 4 and 7b again, in the winding structure of the second electrode **40'**, the uncoated portion of the region from the relative radial position $R_{sub.2,k*+1}$ to a second relative radial position $R_{sub.2,k@}$ of the preset $k@_{sup.th}$ winding turn is also divided into a plurality of segments **61**, the height of the plurality of segments **61** may increase stepwise or progressively toward the outer circumference. Therefore, the region from the relative radial position $R_{sub.2,k*+1}$ to the relative radial position $R_{sub.2,k@}$ corresponds to the height variable region.

(148) In the winding structure of the second electrode **40'**, when the radial length of the height variable region is defined as $H_{sub.2}$ and the ratio of $H_{sub.2}$ to the radius ($R - r_{sub.c}$) of the winding structure of the second electrode **40'** except for the core C is defined as the height variable region ratio ($H_{sub.2}/(R - r_{sub.c})$), the height variable region ratio may be 1% to 56%, like the first electrode.

(149) If the height variable region ratio for the segment **61** of the uncoated portion **43a'** satisfies the above numerical range, the ratio of the radial length of the relative radial positions where the number of stacked layers of the uncoated portion **40** is 10 or more may be at least 30% compared to the radial length ($R_{sub.2}$) of the winding turns including the segment **61**.

(150) In the winding structure of the second electrode **40'**, the uncoated portion of the second electrode **40'** from the second relative radial position $R_{sub.2,k@+1}$ of the preset $k@+1_{sup.th}$ winding turn to the relative radial position **1** is divided into a plurality of segments **61**, and the height of the plurality of segments **61** may be substantially the same from the relative radial position $R_{sub.2,k@+1}$ to the relative radial position **1**.

(151) Preferably, in the winding structure of the first electrode **40**, the uncoated portion **43a** bent toward the core is divided into a plurality of segments **61**, and at least one of a height in a winding axis direction and a width in the winding direction of the plurality of segments **61** may increase gradually or stepwise from the core toward the outer circumference individually or in groups.

(152) Similarly, in the winding structure of the second electrode **40'**, the uncoated portion **43a'** bent

toward the core is divided into a plurality of segments **61**, and at least one of a height in the winding axis direction and a width in the winding direction of the plurality of segments **61** may increase gradually or stepwise from the core toward the outer circumference individually or in groups.

(153) Preferably, when the bending part of the uncoated portions **43a**, **43a'** is divided into a plurality of segments **61**, each of the plurality of segments **61** may satisfy at least one condition among a width (D1 in FIG. 5) condition of 1 mm to 11 mm in the winding direction; a height (D2 in FIG. 5) condition of 2 mm to 10 mm in the winding axis direction; and a separation pitch (D3) condition of 0.05 mm to 1 mm in the winding direction.

(154) Preferably, a predetermined gap may be provided between the bottom portion of the cut groove of the segment **61** (a portion indicated by D4 in FIG. 5) and the active material layer **42**. Preferably, the gap may be 0.2 mm to 4 mm.

(155) Referring to FIG. 4, when the bending part of the uncoated portions **43a**, **43a'** is divided into a plurality of segments **61**, the plurality of segments **61** may form a plurality of segment groups from the core to the outer circumference, and segments belonging to the same segment group may be the same as each other in terms of at least one of the width in the winding direction, the height in the winding axis direction, and the separation pitch in the winding direction.

(156) Preferably, at least a part of the plurality of segment groups may be disposed at the same winding turn of the electrode assembly **80**. In one example, the segments included in each group may constitute at least one winding turn in the winding structure of the electrode assembly **80**. In another example, the segments included in each group may constitute two or more winding turns in the winding structure of the electrode assembly **80**.

(157) FIG. 9a is a partially sectioned view showing an electrode assembly having a radius of 22 mm and included in a cylindrical battery having a form factor of 4680, where the uncoated portion **43a** of the first electrode **40** divided into a plurality of segments **61** is bent from the outer circumference toward the core to form a bending surface region F, in a part of the bending surface region F, the uncoated portion **43a** is overlapped into 10 or more layers along the radial direction, and the stack number increasing region and the stack number uniform region appear along the radial direction of electrode assembly **80**.

(158) Referring to FIG. 9a, the number of stacked layers of the uncoated portion **43a** in the bending surface region F sequentially increases from the outer circumference of the electrode assembly **80** toward the core and reaches a maximum value, and the maximum value is maintained in a predetermined radius region and then decreases by 1 or 2 near the core.

(159) Hereinafter, the radius region in which the number of stacked layers of the uncoated portion **43a** increases sequentially from the outer circumference of the electrode assembly **80** toward the core up to the maximum value is defined as the stack number increasing region, and the region in which the number of stacked layers of the uncoated portion **43a** is maintained at the maximum value and the remaining region near the core are defined as the stack number uniform region together. Since the stack number uniform region includes the region in which the number of stacked layers of the uncoated portion **43a** is maintained at the maximum value, the bending surface region F is flatter than the other regions, which corresponds to an optimal welding region.

(160) In FIG. 9a, the uncoated portion **43a** is divided into segments of a trapezoidal shape as shown in FIG. 5, and only the upper portion of the uncoated portion **43a** is depicted based on the bottom portion **63a** of the cut groove **63**. The uncoated portion **43a** is not shown in a portion corresponding to the cross section of the cut groove **63**.

(161) The points where the segments **61** are actually bent are not exactly the same, and are spaced apart from the lower end of the cut groove **63** by a predetermined distance. As the number of stacked layers of the uncoated portion **43a** increases toward the core, resistance to the overlapping occurs, so it is preferable to perform bending at a point spaced apart from the lower end of the cut groove **63** by a predetermined distance. The separation distance is 2 mm or less, preferably 1 mm

or less. If there is a separation distance, the segments **61** are overlapped better in the radial direction.

(162) The bending surface region F is formed as the segments located at different winding turns are overlapped in the radial direction of the electrode assembly **80**. In the embodiment shown in FIG. **9a**, the segments **61** are not overlapped in the circumferential direction. That is, a gap exists between the sides of the segments **61** as shown in FIG. **6(a)**. The condition of the gap may be satisfied by adjusting width, height, separation pitch, lower internal angle, or the like of the segments. The bending surface region F when the segments are overlapped in the circumferential direction will be described later with reference to FIG. **9b**.

(163) In this embodiment, the radius ($r_{sub.c}$) of the core of the electrode assembly **80** is 4 mm. Also, the height of the segment starts from 3 mm. There is no segment in the uncoated portion **43a** from 4 mm to 7 mm based on the radius of the electrode assembly. That is, segments exist in the region with a radius of 7 mm to 22 mm among the total radius of 22 mm of the electrode assembly, and the width of the radius region where segment **61** exists is 15 mm. If the core is covered by the segment by 10% at maximum based on the radius ($r_{sub.c}$) of the core, the point where segments start to be disposed may be moved toward the core.

(164) In the winding structure, a segment with a height of 3 mm is disposed from the winding turn with a radius of approximately 7 mm. The height of the segment increases from the radius 7 mm of the winding structure by 1 mm per 1 mm increase in radius from the core toward the outer circumference. The period for the height increment of the segment may be changed in the range of 0.2 mm to 1.2 mm per unit radius (1 mm).

(165) FIG. **9a-(a)** is a case where the maximum height of the segment is 8 mm. In this case, the segment is disposed from the point where the radius of the electrode assembly becomes 7 mm from the core center. Only then, when the segment with a height of 3 mm is bent toward the core, the segment does not cover the core with a radius of 4 mm. The height of the segment increases in 5 steps from 3 mm to 8 mm when the radius increases from 7 mm to 12 mm. In addition, the height of the segment is maintained at 8 mm from 12 mm to 22 mm in radius. In this embodiment, the height variable region of the segment is in the radius range of 7 mm to 12 mm, and the height variable region ratio is 28% ($5/18$, rounded to zero decimal place, this will be applied identically below).

(166) FIG. **9a-(b)** is a case where the maximum height of the segment is 7 mm. Also in this case, the segment is disposed from the point where the radius of the electrode assembly becomes 7 mm from the core center. Only then, when the segment with a height of 3 mm is bent toward the core, the segment does not cover the core with a radius of 4 mm. The height of the segment increases in 4 steps from 3 mm to 7 mm when the radius increases from 7 mm to 11 mm. In addition, the height of the segment is maintained at 7 mm from 11 mm to 22 mm in radius. In this embodiment, the height variable region of the segment is in the radius region of 7 mm to 11 mm, and the height variable region ratio is 22% ($4/18$).

(167) FIG. **9a-(c)** is a case where the maximum height of the segment is 6 mm. Also in this case, the segment is disposed from the point where the radius of the electrode assembly becomes 7 mm from the core center. Only then, when the segment with a height of 3 mm is bent toward the core, the segment does not cover the core with a radius of 4 mm. The height of the segment increases in 3 steps from 3 mm to 6 mm when the radius increases from 7 mm to 10 mm. In addition, the height of the segment is maintained at 6 mm from 10 mm to 22 mm in radius. In this embodiment, the height variable region of the segment is in the radius region of 7 mm to 10 mm, and the height variable region ratio is 17% ($3/18$).

(168) In the embodiments shown in (a), (b) and (c) of FIG. **9a**, the height variable region of the segment starts from a radius of 7 mm. In addition, the ratio of the height variable region is 17% to 28%. This ratio range is included in the range of 1% to 56% described above.

(169) Referring to FIG. **9a**, the number of stacked layers of the uncoated portion **43a** sequentially

increases from the outer circumference toward the core. Also, it may be found that even though the minimum length of the segment is the same as 3 mm, the maximum value of the number of stacked layers increases to 12, 15, 18 as the maximum length of the segment increases to 6 mm, 7 mm, and 8 mm. In addition, the thickness of the bending surface region F increases proportionally according to the number of stacked layers.

(170) For example, when the maximum height of the segment is 8 mm, the number of stacked layers of the uncoated portion **43a** increases up to 18 from the outer circumference of the electrode assembly **80** to the core in the 7 mm radius region and, in the 8 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained uniformly at the level of 18. In this example, in the stack number uniform region, the number of stacked layers is at least 16, and its radial width is 8 mm. The width of the stack number uniform region is 53% (8/15, rounded to zero decimal place, this will be applied identically below) compared to the radial length (15 mm) of the winding turns including the segment.

(171) As another example, when the maximum height of the segment is 7 mm, the number of stacked layers of the uncoated portion **43a** increases up to 15 from the outer circumference of the electrode assembly **80** to the core in the 6 mm radius region and, in the 9 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained uniformly at the level of 15. Therefore, the radial width of the stack number uniform region is 9 mm, and the number of stacked layers is at least 13 in the stack number uniform region. The width of the stack number uniform region is 60% (9/15) compared to the radial length (15 mm) of the winding turns including the segment.

(172) As another example, when the maximum height of the segment is 6 mm, the number of stacked layers of the uncoated portion **43a** increases up to 12 from the outer circumference of the electrode assembly **80** to the core in the 5 mm radius region, and in the 10 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained uniformly at the level of 12. Therefore, the radial width of the stack number uniform region is 10 mm, and the number of stacked layers is at least 11 in the stack number uniform region. The width of the stack number uniform region is 67% (10/15) compared to the radial length (15 mm) of the winding turns including the segment.

(173) According to the embodiment, it may be understood that when the minimum length of the segment is 3 mm and the maximum length of the segment is 6 mm, 7 mm and 8 mm, the length of the stack number increasing region in which the number of stacked layers gradually increases is increased to 5 mm, 6 mm and 7 mm, respectively, and the ratio of the stack number uniform region in which the number of stacked layers of the uncoated portion **43a** is 10 or more is 53% to 67%.

(174) Meanwhile, the thickness of the bending surface region F increases in proportion to the number of stacked layers of the uncoated portion **43a**. Depending on the minimum height and the maximum height of the segment in the height variable region, the number of stacked layers of the uncoated portion **43a** may be lowered to 10 and thus the number of stacked layers of the uncoated portion **43a** may be 10 to 18. In one example, when the uncoated portion **43a** is aluminum and its thickness is 10 μm to 25 μm , the thickness of the bending surface region F may be 100 μm to 450 μm . In another example, when the uncoated portion **43a** is copper and its thickness is 5 μm to 20 μm , the thickness of the bending surface region F may be 50 μm to 360 μm . If the thickness of the bending surface region F satisfies the condition of the above numerical range, when the current collector is welded to the bending surface region F using a laser, the bending surface region F absorbs the laser energy sufficiently. As a result, welding beads are formed in a sufficient volume on the bending surface region F to increase the welding strength. In addition, it is possible to prevent the separator or the like located under the bending surface region F from being damaged

since the welding portion is perforated by the laser.

(175) Preferably, the current collector may be welded to the bending surface region F. The welding region of the current collector may at least partially overlap with the stack number uniform region based on the radial direction.

(176) Preferably, 50% to 100% of the welding region of the current collector may overlap with the stack number uniform region in the radial direction of the electrode assembly. As the overlapping ratio of the welding region increases, it is preferable in terms of improving the welding strength and increasing the volume of welding beads. In the welding region of the current collector, the remaining region that does not overlap with the stack number uniform region may overlap with the stack number increasing region.

(177) Meanwhile, as described with reference to FIG. 6, when the segments **61** of the uncoated portion **43a** are bent to form the bending surface region F, if the lower internal angle of the segment included in each segment group satisfies the condition of Formula 1, adjacent segments **61** located at the same winding turn may overlap with each other in the circumferential direction while the sides of the adjacent segments **61** intersect. In this case, the number of stacked layers of the uncoated portion **43a** may be further increased in the radial direction of the electrode assembly.

(178) FIG. **9b** is a cross-sectional view of the bending surface region F exemplarily showing the stack number increasing region and the stack number uniform region when segments are overlapped in the circumferential direction.

(179) Referring to FIG. **9b**, the number of stacked layers of the uncoated portion **43a** sequentially increases from the outer circumference toward the core. The height variable region of the segment starts from the radius of 7 mm as in the embodiment of FIG. **9a**. The height of the segment starts from 3 mm and increases by 1 mm per 1 mm increase in radius. As the maximum value of the segment height is increased to 6 mm, 7 mm, 8 mm, 9 mm and 10 mm, the number of stacked layers at the radial position where the stack number uniform region starts increases to 18, 22, 26, 30 and 34. Under the same conditions where the maximum value of the segment height is 6 mm, 7 mm and 8 mm, the number of stacked layers is greater by 6 to 8 than that of the embodiment of FIG. **9a**. This is because the segments are overlapped in the circumferential direction.

(180) Specifically, when the maximum value of the segment height is 10 mm, the number of stacked layers of the uncoated portion **43a** increases from the outer circumference of the electrode assembly **80** to the core in the 9 mm radius region (the stack number increasing region) up to 34, and, in the 6 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained as 34, and the number of stacked layers further increases to 39 near the core. The number of stacked layers is increased near the core because the segments are overlapped more in the circumferential direction as being closer to the core. In this example, in the stack number uniform region, the number of stacked layers is 34 or more, and its radial width of 6 mm. The stack number uniform region starts from the radius of 7 mm and the stack number uniform region ratio is 40% ($6/15$, rounded to zero decimal place, this will be applied identically below) compared to the radial length (15 mm) of the winding turns including the segment.

(181) As another example, when the maximum value of the segment height is 9 mm, the number of stacked layers of the uncoated portion **43a** increases up to 30 from the outer circumference of the electrode assembly **80** to the core in the 8 mm radius region, and, in the 7 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained as 30, and then further increases to 36 near the core. Therefore, the radial width of the stack number uniform region is 7 mm, and the number of stacked layers is 30 or more in the stack number uniform region. The stack number uniform region starts from the radius of 7 mm and the stack number uniform region ratio is 47% ($7/15$) compared to the radial length (15 mm) of the winding turns including the segment.

(182) As still another example, when the maximum value of the segment height is 8 mm, the

number of stacked layers of the uncoated portion **43a** increases up to 26 from the outer circumference of the electrode assembly **80** to the core in the 7 mm radius region, and, in the 8 mm region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained as 26, and then further increased to 28 near the core. Therefore, the radial width of the stack number uniform region is 8 mm, and the number of stacked layers is 26 or more in the stack number uniform region. The stack number uniform region starts from the radius of 7 mm and the stack number uniform region ratio is 53% (8/15) compared to the radial length (15 mm) of the winding turns including the segment.

(183) As still another example, when the maximum value of the segment height is 7 mm, the number of stacked layers of the uncoated portion **43a** increases up to 22 from the outer circumference of the electrode assembly **80** to the core in the 6 mm radius region, and, in the 9 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained as 22, and then further increased to 23 near the core. Therefore, the radial width of the stack number uniform region is 9 mm, and the number of stacked layers is 22 or more in the stack number uniform region. The stack number uniform region starts from the radius of 7 mm and the stack number uniform region ratio is 60% (9/15) compared to the radial length (15 mm) of the winding turns including the segment.

(184) As still another example, when the maximum value of the segment height is 6 mm, the number of stacked layers of the uncoated portion **43a** increases up to 18 from the outer circumference of the electrode assembly **80** to the core in the 5 mm radius region, and, in the 10 mm radius region toward the core from the radial location where the increasing of the number of stacked layers stops, the number of stacked layers of the uncoated portion **43a** is maintained at **18**, and then further increased to 20 near the core. Therefore, the radial width of the stack number uniform region is 10 mm, and the number of stacked layers is 18 or more in the stack number uniform region. The stack number uniform region starts from the radius of 7 mm and the stack number uniform region ratio is 67% (10/15) compared to the radial length (15 mm) of the winding turns including the segment.

(185) According to the embodiment shown in FIG. **9b**, when the minimum value of the segment height is 3 mm and the maximum value of the segment height is 6 mm, 7 mm, 8 mm, 9 mm and 10 mm, the length of the stack number increasing region in which the number of stacked layers gradually increases is increased to 5 mm, 6 mm, 7 mm, 8 mm and 9 mm. In addition, it may be found that the ratio of the stack number uniform region in which the number of stacked layers is 10 or more is 40% to 67%.

(186) Meanwhile, in the embodiment of FIG. **9b**, the thickness of the bending surface region F increases in proportion to the number of stacked layers of the uncoated portion **43a**. The number of stacked layers of the uncoated portion **43a** is 18 to 39. In one example, when the uncoated portion **43a** is aluminum and its thickness is 10 μm to 25 μm , the thickness of the bending surface region F may be 180 μm to 975 μm . In another example, when the uncoated portion **43a** is copper and its thickness is 5 μm to 20 μm , the thickness of the bending surface region F may be 90 μm to 780 μm . If the thickness of the bending surface region F satisfies the condition of the above numerical range, when the current collector is welded to the bending surface region F using a laser, the bending surface region F absorbs the laser energy sufficiently. As a result, welding beads are formed in a sufficient volume on the bending surface region F to increase the welding strength. In addition, it is possible to prevent the separator and the like located under the bending surface region F from being damaged since the welding portion is perforated by the laser.

(187) Preferably, the welding region of the current collector may at least partially overlap with the stack number uniform region based on the radial direction. Preferably, 50% to 100% of the welding region of the current collector may overlap with the stack number uniform region in the radial

direction of the electrode current collector **80**. As the overlapping ratio of the welding region increases, it is preferable in terms of welding strength. A region of the welding region of the current collector that does not overlap the stack number uniform region may overlap with the stack number increasing region.

(188) In the embodiment illustrated in FIGS. **9a** and **9b**, it will be apparent to those skilled in the art that the stack number uniform region of the uncoated portion **43a** may be increased or decreased according to the radius (R) of the electrode assembly, the radius (r.sub.c) of the core, the minimum and maximum values of the segment height in the segment height variable region, the height increase amount of the segment in the radial direction of the electrode assembly.

(189) The ratio of the stack number uniform region is inversely proportional to the radius of the core (r.sub.c). Also, when the minimum height of the segment is the same, the ratio of the stack number uniform region increases as the radial width of the height variable region decreases. Also, when the maximum height of the segment is the same, the ratio of the stack number uniform region increases as the radial width of the height variable region decreases.

(190) In one example, when the diameter (R) of the electrode assembly is 22 mm, the radius (r.sub.c) of the core is 2 mm and the height of the segment is changed from 7 mm to 10 mm in the radius of 9 mm to 12 mm, which is the height variable region of the segment, the ratio of the stack number uniform region may be decreased to the level of 30%.

(191) In another example, when the diameter (R) of the electrode assembly is 22 mm, the radius (r.sub.c) of the core is 2 mm and the height of the segment is change from 3 mm to 4 mm in the radius of 5 mm to 6 mm, which is the height variable region of the segment, the ratio of the stack number uniform region may be increased to the level of 85%.

(192) Accordingly, the radial length of the stack number uniform region may be 30% or more, preferably 30% to 85%, compared to the radial length of the winding turns including the segment.

(193) Meanwhile, as described by referring to FIGS. **9a** and **9b**, when the maximum height of the segment in the height uniform region of the segment is 6 mm to 10 mm, the number of stacked layers of the uncoated portion **43a** in the stack number uniform region may be adjusted in the range of 10 to 39 by changing the minimum height of segment and the height increment amount of segment in a radial direction. The stack number uniform region of the bending surface region F includes a region formed by bending the segments included in the height uniform region. The thickness of the bending surface region F varies depending on the thickness of the material constituting the uncoated portion **43a**. When the uncoated portion **43a** is made of aluminum and its thickness is 10 μm to 25 μm , the stack thickness of the uncoated portion in the bending surface region F is 100 μm (0.1 mm) to 975 μm (0.975 mm). In this case, the ratio of the stack thickness of the uncoated portion in the bending surface region F to the height of the segment in the bending surface region F formed by bending the segments having height of 6 mm to 10 mm included in the height uniform region is 1.0% (0.1 mm/10 mm) to 16.3% (0.975 mm/6 mm). In another example, when the uncoated portion **43a** is made of copper and its thickness is 5 μm to 20 μm , the stack thickness of the uncoated portion in the bending surface region F is 50 μm (0.05 mm) to 780 μm (0.780 mm). In this case, the ratio of the stack thickness of the uncoated portion in the bending surface region F to the height of the segment in the bending surface region F formed by bending the segments having height of 6 mm to 10 mm included in the height uniform region is 0.5% (0.05 mm/10 mm) to 13.0% (0.780 mm/6 mm). If the thickness ratio of the bending surface region (F) to the height of the segments included in the height uniform region satisfies the above numerical range, the desired welding strength may be achieved when the current collector is welded to the bending surface region F.

(194) Various electrode assembly structures according to the embodiments (modifications) of the present disclosure may be applied to a cylindrical battery or any other batteries well known in the art.

(195) Preferably, the cylindrical battery may be, for example, a cylindrical battery whose form

factor ratio (defined as a value obtained by dividing the diameter of the cylindrical battery by height, namely a ratio of diameter (**1**) to height (H)) is greater than about 0.4.

(196) Here, the form factor means a value indicating the diameter and height of a cylindrical battery. The form factor of the cylindrical battery according to an embodiment of the present disclosure may be, for example, 46110, 4875, 48110, 4880, 4680, or the like. In the numerical value representing the form factor, first two numbers indicate the diameter of the battery, and the remaining numbers indicate the height of the battery.

(197) When an electrode assembly having a tab-less structure is applied to a cylindrical battery having a form factor ratio of more than 0.4, the stress applied in the radial direction when the uncoated portion is bent is large, so that the uncoated portion may be easily torn. In addition, when welding the current collector to the bending surface region of the uncoated portion, it is necessary to sufficiently increase the number of stacked layers of the uncoated portion in order to sufficiently secure the welding strength and lower the resistance. This requirement may be achieved by the electrode and the electrode assembly according to the embodiments (modifications) of the present disclosure.

(198) A battery according to an embodiment of the present disclosure may be a cylindrical battery having an approximately cylindrical shape, whose diameter is approximately 46 mm, height is approximately 110 mm, and form factor ratio is 0.418.

(199) A battery according to another embodiment may be a cylindrical battery having a substantially cylindrical shape, whose diameter is about 48 mm, height is about 75 mm, and form factor ratio is 0.640.

(200) A battery according to still another embodiment may be a cylindrical battery having an approximately cylindrical shape, whose diameter is approximately 48 mm, height is approximately 110 mm, and form factor ratio is 0.436.

(201) A battery according to still another embodiment may be a cylindrical battery having an approximately cylindrical shape, whose diameter is approximately 48 mm, height is approximately 80 mm, and form factor ratio is 0.600.

(202) A battery according to still another embodiment may be a cylindrical battery having an approximately cylindrical shape, whose diameter is approximately 46 mm, height is approximately 80 mm, and form factor ratio is 0.575.

(203) Conventionally, batteries having a form factor ratio of about 0.4 or less have been used. That is, conventionally, for example, 1865 battery, 2170 battery, etc. were used. The 1865 battery has a diameter of approximately 18 mm, height of approximately 65 mm, and a form factor ratio of 0.277. The 2170 battery has a diameter of approximately 21 mm, a height of approximately 70 mm, and a form factor ratio of 0.300.

(204) Hereinafter, the cylindrical battery according to an embodiment of the present disclosure will be described in detail.

(205) FIG. **10** is a sectional view showing a cylindrical battery **190** according to an embodiment of the present disclosure, taken along the Y-axis direction.

(206) Referring to FIG. **10**, the cylindrical battery **190** according to an embodiment of the present disclosure includes an electrode assembly **110** having a first electrode, a separator and a second electrode, a battery housing **142** for accommodating the electrode assembly **110**, and a sealing body **143** for sealing an open end of the battery housing **142**.

(207) The battery housing **142** is a cylindrical container with an opening at the top. The battery housing **142** is made of a conductive metal material such as aluminum or steel. The battery housing **142** accommodates the electrode assembly **110** in the inner space through the top opening and also accommodates the electrolyte.

(208) The electrolyte may be a salt having a structure like A.sup.+B.sup.-. Here, A.sup.+ includes an alkali metal cation such as Li.sup.+, Na.sup.+, or K.sup.+, or a combination thereof and B.sup.- includes at least one anion selected from the group consisting of F.sup.-, Cl.sup.-, Br.sup.-, I.sup.-.

-, No.sub.3.sup.-, N(CN).sub.2.sup.-, BF.sub.4.sup.-, ClO.sub.4.sup.-, AlO.sub.4.sup.-, AlCl.sub.4.sup.-, PF.sub.6.sup.-, SbF.sub.6.sup.-, ASF.sub.6.sup.-, BF.sub.2C.sub.2O.sub.4.sup.-, BC.sub.4O.sub.8.sup.-, (CF.sub.3).sub.2PF.sub.4.sup.-, (CF.sub.3).sub.3PF.sub.3.sup.-, (CF.sub.3).sub.4PF.sub.2.sup.-, (CF.sub.3).sub.5PF.sub.1.sup.-, (CF.sub.3).sub.6P.sub.1.sup.-, CF.sub.3SO.sub.3.sup.-, C.sub.4F.sub.9SO.sub.3.sup.-, CF.sub.3CF.sub.2SO.sub.3.sup.-, (CF.sub.3SO.sub.2).sub.2N.sub.1.sup.-, (FSO.sub.2).sub.2N.sub.1.sup.-, CF.sub.3CF.sub.2(CF.sub.3).sub.2CO.sub.1.sup.-, (CF.sub.3SO.sub.2).sub.2CH.sub.1.sup.-, (SF.sub.5).sub.3C.sub.1.sup.-, (CF.sub.3SO.sub.2).sub.3C.sub.1.sup.-, CF.sub.3(CF.sub.2).sub.7SO.sub.3.sup.-, CF.sub.3CO.sub.2.sup.-, CH.sub.3CO.sub.2.sup.-, SCN.sub.1.sup.- and (CF.sub.3CF.sub.2SO.sub.2).sub.2N.sub.1.sup.-.

(209) The electrolyte may also be dissolved in an organic solvent. The organic solvent may employ propylene carbonate (PC), ethylene carbonate (EC), diethyl carbonate (DEC), dimethyl carbonate (DMC), dipropyl carbonate (DPC), dimethyl sulfoxide, acetonitrile, dimethoxyethane, diethoxyethane, tetrahydrofuran, N-methyl-2-pyrrolidone (NMP), ethyl methyl carbonate (EMC), γ -butyrolactone, or a mixture thereof.

(210) The electrode assembly **110** may have a jelly-roll shape or any other rolled shapes well known in the art. The electrode assembly **110** may be manufactured by winding a laminate formed by sequentially laminating a lower separator, a first electrode, an upper separator, and a second electrode at least once, based on the winding center C, as shown in FIG. 2.

(211) The first electrode and the second electrode have different polarities. That is, if one has positive polarity, the other has negative polarity. At least one of the first electrode and the second electrode may have an electrode structure according to the above embodiments (modifications). In addition, the other of the first electrode and the second electrode may have a conventional electrode structure or an electrode structure according to embodiments (modifications).

(212) An uncoated portion **146a** of the first electrode and an uncoated portion **146b** of the second electrode protrude from the upper and lower portions of the electrode assembly **110**, respectively.

(213) The sealing body **143** may include a cap **143a**, a first gasket **143b** for providing airtightness between the cap **143a** and the battery housing **142** and having insulation, and a connection plate **143c** electrically and mechanically coupled to the cap **143a**.

(214) The cap **143a** is a component made of a conductive metal material, and covers the top opening of the battery housing **142**. The cap **143a** is electrically connected to the uncoated portion **146a** of the first electrode, and is electrically insulated from the battery housing **142** by means of the first gasket **143b**. Accordingly, the cap **143a** may function as a first electrode terminal of the cylindrical battery **140**.

(215) The cap **143a** is placed on the beading portion **147** formed on the battery housing **142**, and is fixed by a crimping portion **148**. Between the cap **143a** and the crimping portion **148**, the first gasket **143b** may be interposed to secure the airtightness of the battery housing **142** and the electrical insulation between the battery housing **142** and the cap **143a**. The cap **143a** may have a protrusion **143d** protruding upward from the center thereof.

(216) The battery housing **142** is electrically connected to the uncoated portion **146b** of the second electrode. Therefore, the battery housing **142** has the same polarity as the second electrode. If the second electrode has negative polarity, the battery housing **142** also has negative polarity.

(217) The battery housing **142** includes the beading portion **147** and the crimping portion **148** at the top thereof. The beading portion **147** is formed by press-fitting the periphery of the outer circumferential surface of the battery housing **142**. The beading portion **147** prevents the electrode assembly **110** accommodated inside the battery housing **142** from escaping through the top opening of the battery housing **142**, and may function as a support portion on which the sealing body **143** is placed.

(218) The crimping portion **148** is formed on the beading portion **147**. The crimping portion **148** has an extended and bent shape to cover the outer circumference of the cap **143a** disposed on the

beading portion **147** and a part of the upper surface of the cap **143a**.

(219) The cylindrical battery **140** may further include a first current collector **144** and/or a second current collector **145** and/or an insulator **146**.

(220) The first current collector **144** is coupled to the upper portion of the electrode assembly **110**. The first current collector **144** is made of a conductive metal material such as aluminum, copper, nickel and so on, and is electrically connected to the bending surface region F.sub.1 that is formed as the uncoated portion **146a** of the first electrode is bent.

(221) A lead **149** may be connected to the first current collector **144**. The lead **149** may extend upward above the electrode assembly **110** and be coupled to the connection plate **143c** or directly coupled to the lower surface of the cap **143a**. The lead **149** may be connected to other components by welding.

(222) Preferably, the first current collector **144** may be integrally formed with the lead **149**. In this case, the lead **149** may have an elongated plate shape extending outward near the center of the first current collector **144**.

(223) The bending surface region (F.sub.1) of the uncoated portion **146a** and the first current collector **144** may be coupled, for example, by laser welding. Laser welding may be performed in a manner that partially melts a base material of the current collector. Laser welding may be replaced by resistance welding, ultrasonic welding, or the like.

(224) Preferably, the uncoated portion **146a** is divided into a plurality of segments, and the bending surface region (F.sub.1) is formed by bending the plurality of segments toward the core C. In the bending surface region (F.sub.1), the radial length of a region where the number of stacked layers of the uncoated portion **146a** is 10 or more may be 30% or more, more particularly, 30% to 85%, compared to the radial length of the winding turns including the segment.

(225) The welding area between the bending surface region (F.sub.1) of the uncoated portion **146a** and the first current collector **144** may overlap with the stack number uniform region (W.sub.1) of the bending surface region (F.sub.1) by 50% or more, and the overlapping ratio may be higher.

(226) When the bending surface region (F.sub.1) of the uncoated portion **146a** and the first current collector **144** are welded with a laser, the welding strength may be 2 kgf/cm² or more, more particularly, 4 kgf/cm² or more. The upper limit of the welding strength may be dependent on a specification of a laser welding equipment. As one example, the welding strength may be set to 8 kgf/cm² or less, or 6 kgf/cm² or less. The laser power for realizing the welding strength varies depending on the laser welding equipment. In one example, the laser power may be in the range of 250 W to 320 W. In another example, the laser power may be adjusted in the range of 40% to 100% of the maximum power specification of the laser welding equipment.

(227) When the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly **110** along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and since the volume of the welding beads is sufficient, the resistance of the welding interface may also be reduced.

(228) The second current collector **145** may be coupled to the lower surface of the electrode assembly **110**. One side of the second current collector **145** may be coupled by welding to the bending surface region (F.sub.2) formed as the uncoated portion **146b** of the second electrode is bent, and the other side may be coupled to the inner bottom surface of the battery housing **142** by welding.

(229) Preferably, the uncoated portion **146b** is divided into a plurality of segments, and the bending surface region (F.sub.2) is formed by bending the plurality of segments toward the core C. In the bending surface region (F.sub.2), the radial length of a region where the number of stacked layers of the uncoated portion **146b** is 10 or more may be 30% or more, more particularly, 30% to 85%, compared to the radial length of the winding turns including the segment.

(230) The coupling structure between the second current collector **145** and the uncoated portion **146b** of the second electrode may be substantially the same as the coupling structure between the

first current collector **144** and the uncoated portion **146a** of the first electrode.

(231) The welding area between the bending surface region (F.sub.2) of the uncoated portion **146b** and the second current collector **145** may overlap with the stack number uniform region (W.sub.2) by 50% or more, and the overlapping ratio may be higher.

(232) When the bending surface region (F.sub.2) of the uncoated portion **146b** and the second current collector **145** are welded with a laser, the welding strength may be 2 kgf/cm.sup.2 or more, more particularly, 4 kgf/cm.sup.2 or more. The upper limit of the welding strength may be dependent on a specification of a laser welding equipment. As one example, the welding strength may be set to 8 kgf/cm.sup.2 or less, or 6 kgf/cm.sup.2 or less. The laser power for realizing the welding strength varies depending on the laser welding equipment. In one example, the laser power may be in the range of 250 W to 320 W. In another example, the laser power may be adjusted in the range of 40% to 100% of the maximum power specification of the laser welding equipment.

(233) When the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly **110** along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and since the volume of the welding beads is sufficient, the resistance of the welding interface may also be reduced.

(234) The insulator **146** may cover the first current collector **144**. The insulator **146** may cover the first current collector **144** at the upper surface of the first current collector **144**, thereby preventing direct contact between the first current collector **144** and the inner circumference of the battery housing **142**.

(235) The insulator **146** has a lead hole **151** so that the lead **149** extending upward from the first current collector **144** may be withdrawn therethrough. The lead **149** is drawn upward through the lead hole **151** and coupled to the lower surface of the connection plate **143c** or the lower surface of the cap **143a**.

(236) A peripheral region of the edge of the insulator **146** may be interposed between the first current collector **144** and the beading portion **147** to fix the coupled body of the electrode assembly **110** and the first current collector **144**. Accordingly, the movement of the coupled body of the electrode assembly **110** and the first current collector **144** may be restricted in the height direction of the battery **140**, thereby improving the assembly stability of the battery **140**.

(237) The insulator **146** may be made of an insulating polymer resin. In one example, the insulator **146** may be made of polyethylene, polypropylene, polyimide, or polybutylene terephthalate.

(238) The battery housing **142** may further include a venting portion **152** formed at a lower surface thereof. The venting portion **152** corresponds to a region having a smaller thickness compared to the peripheral region of the lower surface of the battery housing **142**. The venting portion **152** is structurally weak compared to the surrounding area. Accordingly, when an abnormality occurs in the cylindrical battery **190** and the internal pressure increases to a predetermined level or more, the venting portion **152** may be ruptured so that the gas generated inside the battery housing **142** is discharged to the outside.

(239) The venting portion **152** may be formed continuously or discontinuously while drawing a circle at the lower surface of the battery housing **142**. In a modification, the venting portion **152** may be formed in a straight pattern or other patterns.

(240) FIG. **11** is a sectional view showing a cylindrical battery **200** according to still another embodiment of the present disclosure, taken along the Y-axis.

(241) Referring to FIG. **11**, the structure of the electrode assembly of the cylindrical battery **200** is substantially the same as that of the cylindrical battery **190** of in FIG. **10**, and the other structure except for the electrode assembly is changed.

(242) Specifically, the cylindrical battery **200** includes a battery housing **171** through which a terminal **172** is installed. The terminal **172** is installed on the closed surface (the upper surface in the drawing) of the battery housing **171**. The terminal **172** is riveted to a perforation hole of the battery housing **171** in a state where a second gasket **173** made of an insulating material is

interposed therebetween. The terminal **172** is exposed to the outside in a direction opposite to the direction of gravity.

(243) The terminal **172** includes a terminal exposing portion **172a** and a terminal insert portion **172b**. The terminal exposing portion **172a** is exposed to the outside of the closed surface of the battery housing **171**. The terminal exposing portion **172a** may be located approximately at a central portion of the closed surface of the battery housing **171**. The maximum diameter of the terminal exposing portion **172a** may be larger than the maximum diameter of the perforation hole formed in the battery housing **171**. The terminal insert portion **172b** may be electrically connected to the uncoated portion **146a** of the first electrode through approximately the central portion of the closed surface of the battery housing **171**. The terminal insert portion **172b** may be riveted onto the inner surface of the battery housing **171**. That is, the lower edge of the terminal insert portion **172b** may have a shape curved toward the inner surface of the battery housing **171**. The maximum diameter of the end of the terminal insert portion **172b** may be larger than the maximum diameter of the perforation hole of the battery housing **171**.

(244) The lower surface of the terminal insert portion **172b** is substantially flat and may be welded to the center portion of the first current collector **144** connected to the uncoated portion **146a** of the first electrode. An insulator **174** made of an insulating material may be interposed between the first current collector **144** and the inner surface of the battery housing **171**. The insulator **174** covers the upper portion of the first current collector **144** and the top edge of the electrode assembly **110**. Accordingly, it is possible to prevent the uncoated portion **146a** exposed at the outer circumference of the electrode assembly **110** from contacting the inner surface of the battery housing **171** having a different polarity to cause a short circuit.

(245) The insulator **174** is in contact with the inner surface of the closed portion of the battery housing **171** and in contact with the upper surface of the first current collector **144**. To this end, the insulator **174** has a thickness corresponding to the separation distance between the inner surface of the closed portion of the battery housing **171** and the upper surface of the first current collector **144**, or a thickness slightly larger than the separation distance.

(246) Preferably, the first current collector **144** may be laser-welded to the bending surface region F.sub.1 of the uncoated portion **146a**. At this time, welding is performed in a region including the stack number uniform region where the number of stacked layers of the uncoated portion **146a** is 10 or more in the bending surface region F.sub.1 of the uncoated portion **146a**.

(247) The radial length of the stack number uniform region in which the number of stacked layers of the uncoated portion **146a** is 10 or more may be 30% or more, more particularly, 30% to 85%, compared to the radial length of the winding turns including the segment.

(248) The welding area between the bending surface region (F.sub.1) of the uncoated portion **146a** and the first current collector **144** may overlap with the stack number uniform region (W.sub.1) by 50% or more, and the overlapping ratio may be higher.

(249) When the bending surface region (F.sub.1) of the uncoated portion **146a** and the first current collector **144** are welded with a laser, the welding strength may be 2 kgf/cm^{sup.2} or more, more particularly, 4 kgf/cm^{sup.2} or more. The upper limit of the welding strength may be dependent on a specification of a laser welding equipment. As one example, the welding strength may be set to 8 kgf/cm^{sup.2} or less, or 6 kgf/cm^{sup.2} or less. The laser power for realizing the welding strength varies depending on the laser welding equipment. In one example, the laser power may be in the range of 250 W to 320 W. In another example, the laser power may be adjusted in the range of 40% to 100% of the maximum power specification of the laser welding equipment.

(250) When the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly **110** along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and since the volume of the welding beads is sufficient, the resistance of the welding interface may also be reduced.

(251) The second gasket **173** is interposed between the battery housing **171** and the terminal **172** to

prevent the battery housing **171** and the terminal **172** having opposite polarities from electrically contacting each other. Accordingly, the upper surface of the battery housing **171** having an approximately flat shape may function as a second electrode terminal of the cylindrical battery **200**. (252) The second gasket **173** includes a gasket exposing portion **173a** and a gasket insert portion **173b**. The gasket exposing portion **173a** is interposed between the terminal exposing portion **172a** of the terminal **172** and the battery housing **171**. The gasket insert portion **173b** is interposed between the terminal insert portion **172b** of the terminal **172** and the battery housing **171**. The gasket insert portion **173b** may be deformed together when the terminal insert portion **172b** is riveted, so as to be in close contact with the inner surface of the battery housing **171**. The second gasket **173** may be made of, for example, a polymer resin having insulation.

(253) The gasket exposing portion **173a** of the second gasket **173** may have an extended shape to cover the outer circumference of the terminal exposing portion **172a** of the terminal **172**. When the second gasket **173** covers the outer circumference of the terminal **172**, it is possible to prevent a short circuit from occurring while an electrical connection part such as a bus bar is coupled to the upper surface of the battery housing **171** and/or the terminal **172**. Although not shown in the drawings, the gasket exposing portion **173a** may have an extended shape to cover not only the outer circumference surface of the terminal exposing portion **172a** but also a part of the upper surface thereof.

(254) When the second gasket **173** is made of a polymer resin, the second gasket **173** may be coupled to the battery housing **171** and the terminal **172** by thermal fusion. In this case, airtightness at the coupling interface between the second gasket **173** and the terminal **172** and at the coupling interface between the second gasket **173** and the battery housing **171** may be enhanced. Meanwhile, when the gasket exposing portion **173a** of the second gasket **173** has a shape extending to the upper surface of the terminal exposing portion **172a**, the terminal **172** may be integrally coupled with the second gasket **173** by insert injection molding.

(255) In the upper surface of the battery housing **171**, a remaining area **175** other than the area occupied by the terminal **172** and the second gasket **173** corresponds to the second electrode terminal having a polarity opposite to that of the terminal **172**.

(256) The second current collector **176** is coupled to the lower portion of the electrode assembly **110**. The second current collector **176** is made of a conductive metal material such as aluminum, steel, copper or nickel, and is electrically connected to the uncoated portion **146b** of the second electrode.

(257) Preferably, the second current collector **176** is electrically connected to the battery housing **171**. To this end, at least a portion of the edge of the second current collector **176** may be interposed and fixed between the inner surface of the battery housing **171** and a first gasket **178b**.

(258) In one example, at least a portion of the edge of the second current collector **176** may be fixed to the beading portion **180** by welding in a state of being supported on the lower surface of the beading portion **180** formed at the bottom of the battery housing **171**. In a modification, at least a portion of the edge of the second current collector **176** may be directly welded to the inner wall surface of the battery housing **171**.

(259) Preferably, the second current collector **176** and the bending surface region (F.sub.2) of the uncoated portion **146b** may be coupled by welding, for example, laser welding. At this time, welding is performed in an area including a stack number uniform region where the number of stacked layers of the uncoated portion **146b** is 10 or more in the bending surface region (F.sub.2) of the uncoated portion **146b**.

(260) The radial length of the region where the number of stacked layers of the uncoated portion **146b** is 10 or more may be 30% or more, more particularly, 30% to 85%, compared to the radial length of the winding turns including the segment.

(261) The welding area between the bending surface region (F.sub.2) of the uncoated portion **146b** and the second current collector **176** may overlap with the stack number uniform region (W.sub.2)

by 50% or more, and the overlapping ratio may be higher.

(262) When the bending surface region (F.sub.2) of the uncoated portion **146b** and the second current collector **176** are welded with a laser, the welding strength may be 2 kgf/cm.sup.2 or more, more particularly, 4 kgf/cm.sup.2 or more.

(263) When the welding strength satisfies the above numerical range, even if severe vibration is applied to the electrode assembly **110** along the winding axis direction and/or the radial direction, the properties of the welding interface do not deteriorate, and since the volume of the welding beads is sufficient, the resistance of the welding interface may also be reduced.

(264) A sealing body **178** for sealing the lower open end of the battery housing **171** includes a cap **178a** and a first gasket **178b**. The first gasket **178b** electrically separates the cap **178a** and the battery housing **171**. A crimping portion **181** fixes the edge of the cap **178a** and the first gasket **178b** together. The cap **178a** has a venting portion **179**. The configuration of the venting portion **179** is substantially the same as the above embodiment (modification).

(265) Preferably, the cap **178a** is made of a conductive metal material. However, since the first gasket **178b** is interposed between the cap **178a** and the battery housing **171**, the cap **178a** does not have electrical polarity. The sealing body **178** seals the open end of the lower portion of the battery housing **171** and functions to discharge gas when the internal pressure of the battery **200** increases over a critical value.

(266) Preferably, the terminal **172** electrically connected to the uncoated portion **146a** of the first electrode is used as the first electrode terminal. In addition, in the upper surface of the battery housing **171** electrically connected to the uncoated portion **146b** of the second electrode through the second current collector **176**, a part **175** except for the terminal **172** is used as the second electrode terminal having a different polarity from the first electrode terminal. If two electrode terminals are located at the upper portion of the cylindrical battery **200** as above, it is possible to arrange electrical connection components such as bus bars at only one side of the cylindrical battery **200**. This may bring about simplification of the battery pack structure and improvement of energy density. In addition, since the part **175** used as the second electrode terminal has an approximately flat shape, a sufficient bonding area may be secured for bonding electrical connection components such as bus bars. Accordingly, the cylindrical battery **200** may reduce the resistance at the bonding portion of the electrical connection components to a desirable level.

(267) In the present disclosure, even when the uncoated portions **146a**, **146b** are bent toward the core, the core C of the electrode assembly **110** may be opened upward without being blocked.

(268) That is, as shown in FIG. 4, the height of the uncoated portion of the first and second electrodes, particularly the height of the core-side uncoated portion A, is designed to be low, and the height variable region of the segment **61** is disposed adjacent to the core-side uncoated portion A, so that by adjusting the height of the segment **61** closest to the core-side uncoated portion A, the core C of the electrode assembly **110** is not blocked even if the uncoated portion near the core of the electrode assembly **110** is bent.

(269) If the core C is not blocked, there is no difficulty in the electrolyte injection process and the electrolyte injection efficiency is improved. In addition, by inserting a welding jig through the core C, the welding process between the current collector **145** and the bottom of the battery housing **142** or the welding process between the current collector **144** and the terminal **172** may be easily performed.

(270) When the uncoated portions **146a**, **146b** have a segment structure, if the width and/or height and/or separation pitch of the segments are adjusted to satisfy the numerical ranges of the above embodiment, the segments are overlapped in multiple layers to sufficiently secure welding strength when the segments are bent, and an empty space (gap) is not formed on the bending surface region (F.sub.1, F.sub.2).

(271) Meanwhile, the first current collector **144** and the second current collector **176** may have a new structure as shown in FIGS. 12 and 13.

(272) FIG. 12 is a top plan view showing the structure of the first current collector **144** according to an embodiment of the present disclosure.

(273) Referring to FIG. 12, the first current collector **144** may include an edge portion **144a**, a first uncoated portion coupling portion **144b**, and a terminal coupling portion **144c**. The edge portion **144a** is disposed on the electrode assembly **110**. The edge portion **144a** may have a substantially rim shape having an empty space S formed therein. In the drawings of the present disclosure, only a case in which the edge portion **144a** has a substantially circular rim shape is illustrated, but the present disclosure is not limited thereto. The edge portion **144a** may have a substantially rectangular rim shape, a hexagonal rim shape, an octagonal rim shape, or other rim shapes, unlike the illustrated one.

(274) The terminal coupling portion **144c** may have a diameter equal to or greater than the diameter of the flat portion formed on the bottom surface of the terminal **172** in order to secure a welding area for coupling with the flat portion formed on the bottom surface of the terminal **172**.

(275) The first uncoated portion coupling portion **144b** extends inward from the edge portion **144a** and is coupled to the uncoated portion **146a**. The terminal coupling portion **144c** is spaced apart from the first uncoated portion coupling portion **144b** and is positioned inside the edge portion **144a**. The terminal coupling portion **144c** may be coupled to the terminal **172** by welding. The terminal coupling portion **144c** may be located, for example, approximately at the center of the inner space surrounded by the edge portion **144a**. The terminal coupling portion **144c** may be provided at a position corresponding to the hole formed in the core C of the electrode assembly **110**. The terminal coupling portion **144c** may be configured to cover the hole formed in the core C of the electrode assembly **110** so that the hole formed in the core C of the electrode assembly **110** is not exposed out of the terminal coupling portion **144c**. To this end, the terminal coupling portion **144c** may have a larger diameter or width than the hole formed in the core C of the electrode assembly **110**.

(276) The first uncoated portion coupling portion **144b** and the terminal coupling portion **144c** may not be directly connected, but may be disposed to be spaced apart from each other and indirectly connected by the edge portion **144a**. Since the first current collector **144** has a structure in which the first uncoated portion coupling portion **144b** and the terminal coupling portion **144c** are not directly connected to each other but are connected through the edge portion **144c** as above, when shock and/or vibration occurs at the cylindrical battery **200**, it is possible to disperse the shock applied to the coupling portion between the first uncoated portion coupling portion **144b** and the first uncoated portion **146a** and the coupling portion between the terminal coupling portion **144c** and the terminal **172**. In the drawings of the present disclosure, only a case in which four first uncoated portion coupling portions **144b** are provided is illustrated, but the present disclosure is not limited thereto. The number of the first uncoated portion coupling portions **144b** may be variously determined in consideration of manufacturing difficulty according to the complexity of the shape, electric resistance, the space inside the edge portion **144a** considering electrolyte impregnation, and the like.

(277) The first current collector **144** may further include a bridge portion **144d** extending inward from the edge portion **144a** and connected to the terminal coupling portion **144c**. At least a part of the bridge portion **144d** may have a smaller sectional area compared to the first uncoated portion coupling portion **144b** and the edge portion **144a**. For example, at least a part of the bridge portion **144d** may be formed to have a smaller width and/or thickness compared to the first uncoated portion coupling portion **144b**. In this case, the electric resistance increases in the bridge portion **144d**, and thus, when a current flows through the bridge portion **144d**, the relatively large resistance causes a part of the bridge portion **144d** to be melted due to overcurrent heating, thereby irreversibly blocking the overcurrent. The sectional area of the bridge portion **144d** may be adjusted to an appropriate level in consideration of the overcurrent blocking function.

(278) The bridge portion **144d** may include a taper portion **144e** whose width is gradually

decreased from the inner surface of the edge portion **144a** toward the terminal coupling portion **144c**. When the taper portion **144e** is provided, the rigidity of the component may be improved at the connection portion between the bridge portion **144d** and the edge portion **144a**. When the taper portion **144e** is provided, in the process of manufacturing the cylindrical battery **200**, for example, a transfer device and/or a worker may easily and safely transport the first current collector **144** and/or a coupled body of the first current collector **144** and the electrode assembly **110** by gripping the taper portion **144e**. That is, when the taper portion **144e** is provided, it is possible to prevent product defects that may occur by gripping a portion where welding is performed with other components such as the first uncoated portion coupling portion **144b** and the terminal coupling portion **144c**.

(279) The first uncoated portion coupling portion **144b** may be provided in plural. The plurality of first uncoated portion coupling portions **144b** may be disposed substantially at regular intervals from each other in the extending direction of the edge portion **144a**. An extension length of each of the plurality of first uncoated portion coupling portions **144b** may be substantially equal to each other. The first uncoated portion coupling portion **144b** may be coupled to the bending surface region (F.sub.1) of the uncoated portion **146a** by laser welding. The welding pattern **144f** formed by welding between the first uncoated portion coupling portion **144b** and the bending surface region (F.sub.1) may have a structure to extend along the radial direction of the electrode assembly **110**. The welding pattern **144f** may be a line pattern or a dot array pattern.

(280) The terminal coupling portion **144c** may be disposed to be surrounded by the plurality of first uncoated portion coupling portions **144b**. The terminal coupling portion **144c** may be coupled to the terminal **172** by welding. The bridge portion **144d** may be positioned between a pair of first uncoated portion coupling portions **144b** adjacent to each other. In this case, the distance from the bridge portion **144d** to any one of the pair of first uncoated portion coupling portions **144b** along the extending direction of the edge portion **144a** may be substantially equal to the distance from the bridge portion **144d** to the other one of the pair of first uncoated portion coupling portions **144b** along the extending direction of the edge portion **144a**. The plurality of first uncoated portion coupling portions **144b** may be formed to have substantially the same sectional area. The plurality of first uncoated portion coupling portions **144b** may be formed to have substantially the same width and thickness.

(281) Although not shown in the drawings, the bridge portion **144d** may be provided in plural. Each of the plurality of bridge portions **144d** may be disposed between a pair of first uncoated portion coupling portions **144b** adjacent to each other. The plurality of bridge portions **144d** may be disposed substantially at regular intervals to each other in the extending direction of the edge portion **144a**. A distance from each of the plurality of bridge portions **144d** to one of the pair of first uncoated portion coupling portions **144b** adjacent to each other along the extending direction of the edge portion **144a** may be substantially equal to a distance from each of the plurality of the bridge portion **144d** to the other first uncoated portion coupling portion **144b**.

(282) In the case where the first uncoated portion coupling portion **144b** and/or the bridge portion **144d** is provided in plural as described above, if the distance between the first uncoated portion coupling portions **144b** and/or the distance between the bridge portions **144d** and/or the distance between the first uncoated portion coupling portion **144b** and the bridge portion **144d** is uniformly formed, a current flowing from the first uncoated portion coupling portion **144b** toward the bridge portion **144d** or a current flowing from the bridge portion **144d** toward the first uncoated portion coupling portion **144b** may be smoothly and uniformly formed.

(283) Meanwhile, the first current collector **144** and the bending surface region (F.sub.1) of the uncoated portion **146a** may be coupled by welding. In this case, laser welding, ultrasonic welding, spot welding, or the like may be applied, for example. Preferably, the welding region may overlap with the stack number uniform region (W.sub.1) of the bending surface region (F.sub.1) by 50% or more.

(284) The bridge portion **144d** may include a notching portion N formed to partially reduce a sectional area of the bridge portion **144d**. The sectional area of the notching portion N may be adjusted, for example, by partially reducing the width and/or thickness of the bridge portion **144d**. When the notching portion N is provided, electric resistance is increased in the region where the notching portion N is formed, thereby enabling rapid current interruption when overcurrent occurs.

(285) The notching portion N may be provided in a region corresponding to the overlapping layer number uniform region of the electrode assembly **110** in order to prevent foreign substances generated during rupturing from flowing into the electrode assembly **110**. This is because, in this region, the number of stacked layers of the segments of the uncoated portion **146a** is maintained to the maximum and thus the overlapped segments may function as a mask. For example, the notching portion N may be provided in a region in which the number of stacked layers of the uncoated portion **146a** is maximum in the stack number uniform region.

(286) FIG. **13** is a perspective view showing the structure of the second current collector **176** according to an embodiment of the present disclosure.

(287) Referring to FIG. **13**, the second current collector **176** is disposed below the electrode assembly **110**. In addition, the second current collector **176** may be configured to electrically connect the uncoated portion **146b** of the electrode assembly **110** and the battery housing **171**. The second current collector **176** is made of a metal material with conductivity and is electrically connected to the uncoated portion **146b**. In addition, the second current collector **176** is electrically connected to the battery housing **171**. The second current collector **176** may be interposed and fixed between the inner surface of the battery housing **171** and the first gasket **178b**. Specifically, the second current collector **176** may be interposed between the lower surface of the beading portion **180** of the battery housing **171** and the first gasket **178b**. However, the present disclosure is not limited thereto, and the second current collector **176** may be welded to the inner wall surface of the battery housing **171** in a region where the beading portion **180** is not formed.

(288) The second current collector **176** may include a support portion **176a** disposed below the electrode assembly **110**, a second uncoated portion coupling portion **176b** extending from the support portion **176a** approximately along the radial direction of the electrode assembly **110** and coupled to the bending surface region (F.sub.2) of the uncoated portion **146b**, and a housing coupling portion **176c** extending from the support portion **176a** approximately along the radial direction of the electrode assembly **110** and coupled to the inner surface of the battery housing **171**. The second uncoated portion coupling portion **176b** and the housing coupling portion **176c** are indirectly connected through the support portion **176a**, and are not directly connected to each other. Therefore, when an external shock is applied to the cylindrical battery **200** of the present disclosure, it is possible to minimize the possibility of damage to the coupling portion of the second current collector **176** and the electrode assembly **110** and the coupling portion of the second current collector **176** and the battery housing **171**. However, the second current collector **176** of the present disclosure is not limited to the structure where the second uncoated portion coupling portion **176b** and the housing coupling portion **176c** are only indirectly connected. For example, the second current collector **176** may have a structure that does not include the support portion **176a** for indirectly connecting the second uncoated portion coupling portion **176b** and the housing coupling portion **176c** and/or a structure in which the uncoated portion **146b** and the housing coupling portion **176c** are directly connected to each other.

(289) The support portion **176a** and the second uncoated portion coupling portion **176b** are disposed below the electrode assembly **110**. The second uncoated portion coupling portion **176b** is coupled to the bending surface region (F.sub.2) of the uncoated portion **146b**. In addition to the second uncoated portion coupling portion **176b**, the support portion **176a** may also be coupled to the uncoated portion **146b**. The second uncoated portion coupling portion **176b** and the uncoated portion **146b** may be coupled by welding. The support portion **176a** and the second uncoated portion coupling portion **176b** are located higher than the beading portion **180** when the beading

portion **180** is formed on the battery housing **171**.

(290) The support portion **176a** has a current collector hole **176d** formed at a location corresponding to the hole formed at the core C of the electrode assembly **110**. The core C of the electrode assembly **110** and the current collector hole **176d** communicating with each other may function as a passage for inserting a welding rod for welding between the terminal **172** and the terminal coupling portion **144c** of the first current collector **144** or for irradiating a laser beam. The current collector hole **176d** may have a diameter substantially equal to or greater than the hole formed in the core C of the electrode assembly **110**. When the second uncoated portion coupling portion **176b** is provided in plural, the plurality of second uncoated portion coupling portions **176b** may have a shape extending approximately radially from the support portion **176a** of the second current collector **176** toward the sidewall of the battery housing **171**. The plurality of second uncoated portion coupling portions **176b** may be positioned to be spaced apart from each other along the periphery of the support portion **176a**.

(291) The housing coupling portion **176c** may be provided in plural. In this case, the plurality of housing coupling portions **176c** may have a shape extending approximately radially from the center of the second current collector **176** toward the sidewall of the battery housing **171**. Accordingly, the electrical connection between the second current collector **176** and the battery housing **171** may be made at a plurality of points. Since the coupling for electrical connection is made at a plurality of points, the coupling area may be maximized, thereby minimizing electric resistance. The plurality of housing coupling portions **176c** may be positioned to be spaced apart from each other along the periphery of the support portion **176a**. At least one housing coupling portion **176c** may be positioned between the second uncoated portion coupling portions **176b** adjacent to each other. The plurality of housing coupling portions **176c** may be coupled to, for example, the beading portion **180** in the inner surface of the battery housing **171**. The housing coupling portions **176c** may be coupled, particularly, to the lower surface of the beading portion **180** by welding. The welding may employ, for example, laser welding, ultrasonic welding, or spot welding. By coupling the housing coupling portions **176c** on the beading portion **180** by welding in this way, the resistance level of the cylindrical battery **200** may be limited to about 4 milliohms or less. In addition, as the lower surface of the beading portion **180** has a shape extending in a direction approximately parallel to the upper surface of the battery housing **171**, namely in a direction approximately perpendicular to the sidewall of the battery housing **171**, and the housing coupling portion **176c** also has a shape extending in the same direction, namely in the radial direction and the circumferential direction, the housing coupling portion **176c** may be stably in contact with the beading portion **180**. In addition, as the housing coupling portion **176c** is stably in contact with the flat portion of the beading portion **180**, the two components may be welded smoothly, thereby improving the coupling force between the two components and minimizing the increase in resistance at the coupling portion.

(292) The housing coupling portion **176c** may include a contact portion **176e** coupled onto the inner surface of the battery housing **171** and a connection portion **176f** for connecting the support portion **176a** and the contact portion **176e**.

(293) The contact portion **176e** is coupled onto the inner surface of the battery housing **171**. In the case where the beading portion **180** is formed on the battery housing **171**, the contact portion **176e** may be coupled onto the beading portion **180** as described above. More specifically, the contact portion **176e** may be electrically coupled to the flat portion formed at the lower surface of the beading portion **180** formed on the battery housing **171**, and may be interposed between the lower surface of the beading portion **180** and the first gasket **178b**. In this case, for stable contact and coupling, the contact portion **176e** may have a shape extending from the beading portion **180** by a predetermined length along the circumferential direction of the battery housing **171**.

(294) Meanwhile, the maximum distance from the center of the second current collector **176** to the end of the second uncoated portion coupling portion **176b** along the radial direction of the electrode assembly **110** may be equal to or smaller than the inner diameter of the battery housing **171** in a

region where the beading portion **180** is formed, namely the minimum inner diameter of the battery housing **171**. This is to prevent the second current collector **176** from being interfered by the beading portion **180** during the sizing process of compressing the battery housing **171** along the height direction, and thus to prevent the electrode assembly **110** from being pressed by the second current collector **176**.

(295) The second uncoated portion coupling portion **176b** includes a hole **176g**. The hole **176g** may be used as a passage through which the electrolyte may move. The welding pattern **176h** formed by welding between the second uncoated portion coupling portion **176b** and the bending surface region (F.sub.2) may have a structure to extend along the radial direction of the electrode assembly **110**. The welding pattern **176h** may be a line pattern or a dot array pattern.

(296) The cylindrical battery **200** according to an embodiment of the present disclosure have an advantage in that electrical connection can be performed at the upper portion thereof.

(297) FIG. **14** is a top plan view illustrating a state in which a plurality of cylindrical batteries **200** are electrically connected, and FIG. **15** is a partially enlarged view of FIG. **14**.

(298) Referring to FIGS. **14** and **15**, a plurality of cylindrical batteries **200** may be connected in series and in parallel at an upper portion of the cylindrical batteries **200** using a bus bar **210**. The number of cylindrical batteries **200** may be increased or decreased in consideration of the capacity of the battery pack.

(299) In each cylindrical battery **200**, the terminal **172** may have a positive polarity, and the flat surface **171a** around the terminal **172** of the battery housing **171** may have a negative polarity, or vice versa.

(300) Preferably, the plurality of cylindrical batteries **200** may be arranged in a plurality of columns and rows. Columns are provided in a vertical direction with respect to the drawing, and rows are provided in a left and right direction with respect to the drawing. In addition, in order to maximize space efficiency, the cylindrical batteries **200** may be arranged in a closest packing structure. The closest packing structure is formed when an equilateral triangle is formed by connecting the centers of the terminals **172** exposed out of the battery housing **171** to each other. Preferably, the bus bar **210** connects the cylindrical batteries **200** arranged in the same column in parallel to each other, and connects the cylindrical batteries **200** arranged in two neighboring columns in series with each other.

(301) Preferably, the bus bar **210** may include a body portion **211**, a plurality of first bus bar terminals **212** and a plurality of second bus bar terminals **213** for serial and parallel connection. The body portion **211** may extend along the column of the cylindrical battery **200** between neighboring terminals **172**. Alternatively, the body portion **211** may extend along the row of the cylindrical batteries **1** and may be regularly bent like a zigzag shape.

(302) The plurality of first bus bar terminals **212** may extend from one side of the body portion **211** and may be electrically coupled to the terminal **172** of the cylindrical battery **200** located in the extending direction. The electrical connection between the first bus bar terminal **212** and the terminal **172** may be achieved by laser welding, ultrasonic welding, or the like.

(303) The plurality of second bus bar terminals **213** may extend from the other side of the body portion **211** and may be electrically coupled to the flat surface **171a** around the terminal **172** located in the extending direction. The electrical coupling between the second bus bar terminal **213** and the flat surface **171a** may be performed by laser welding, ultrasonic welding, or the like.

(304) Preferably, the body portion **211**, the plurality of first bus bar terminals **212** and the plurality of second bus bar terminals **213** may be made of one conductive metal plate. The metal plate may be, for example, an aluminum plate or a copper plate, but the present disclosure is not limited thereto. In a modified example, the body portion **211**, the plurality of first bus bar terminals **212** and the second bus bar terminals **213** may be manufactured as separate pieces and then coupled to each other by welding or the like.

(305) The cylindrical battery **200** of the present disclosure as described above has a structure in

which resistance is minimized by enlarging the welding area by means of the bending surface region F.sub.1 and F.sub.2, multiplexing current paths by means of the second current collector **176**, minimizing a current path length, or the like. The AC resistance of the cylindrical battery **200** measured through a resistance meter between the positive electrode and the negative electrode, namely between the terminal **172** and the flat surface **171a** around the terminal **172**, may be about 4 milliohms or below, but greater than 0 milliohms, such as 0.01 milliohms, suitable for fast charging.

(306) In the cylindrical battery **200** according to the present disclosure, since the terminal **172** having a positive polarity and the flat surface **171a** having a negative polarity are located in the same direction, it is easy to electrically connect the cylindrical batteries **200** using the bus bar **210**.

(307) In addition, since the terminal **172** of the cylindrical battery **200** and the flat surface **171a** around the terminal **172** have a large area, the coupling area of the bus bar **210** may be sufficiently secured to sufficiently reduce the resistance of the battery pack including the cylindrical battery **200**.

(308) The cylindrical battery according to the above embodiments (modifications) may be used to manufacture a battery pack.

(309) FIG. **16** is a diagram schematically showing a battery pack according to an embodiment of the present disclosure.

(310) Referring to FIG. **16**, a battery pack **300** according to an embodiment of the present disclosure includes an aggregate in which cylindrical batteries **301** are electrically connected, and a pack housing **302** for accommodating the aggregate. The cylindrical battery **301** may be any one of the batteries according to the above embodiments (modifications). In the drawing, components such as a bus bar, a cooling unit, and an external terminal for electrical connection of the cylindrical batteries **301** are not depicted for convenience of illustration.

(311) The battery pack **300** may be mounted to a vehicle. The vehicle may be, for example, an electric vehicle, a hybrid electric vehicle, or a plug-in hybrid vehicle. The vehicle includes a four-wheeled vehicle or a two-wheeled vehicle.

(312) FIG. **17** is a diagram schematically showing a vehicle including the battery pack **300** of FIG. **16**.

(313) Referring to FIG. **17**, a vehicle V according to an embodiment of the present disclosure includes the battery pack **300** according to an embodiment of the present disclosure. The vehicle V operates by receiving power from the battery pack **300** according to an embodiment of the present disclosure.

(314) According to an embodiment of the present disclosure, when bending the uncoated portions exposed at both ends of the electrode assembly, it is possible to prevent the separator or the active material layer from being damaged when welding the current collector by sufficiently securing an area where the uncoated portion is overlapped into 10 or more layers in the radial direction of the electrode assembly.

(315) According to still another embodiment of the present disclosure, since the structure of the uncoated portion adjacent to the core of the electrode assembly is improved, it is possible to prevent the cavity in the core of the electrode assembly from being blocked when the uncoated portion is bent. Thus, the electrolyte injection process and the process of welding the battery housing and the current collector may be carried out easily.

(316) According to still another embodiment of the present disclosure, since the bending surface region of the uncoated portion is directly welded to the current collector instead of a strip-shaped electrode tab, it is possible to provide an electrode assembly with improved energy density and reduced resistance.

(317) According to still another embodiment of the present disclosure, it is possible to provide a cylindrical battery having a structure that has a low internal resistance and improves welding strength between the current collector and the uncoated portion, and a battery pack and a vehicle

including the cylindrical battery.

(318) The present disclosure has been described in detail. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the scope of the disclosure will become apparent to those skilled in the art from this detailed description.

Claims

1. An electrode assembly, comprising: a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference of the electrode assembly, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region including stacked layers of the second portion, wherein, in a partial region of the first surface region, a number of the stacked layers of the second portion is 10 or more in the first axial direction, wherein the second portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly.

2. The electrode assembly according to claim 1, wherein a number of total winding turns of the first electrode is defined as $n_{\text{sub.1}}$, wherein a relative radial position $R_{\text{sub.1},k}$ is defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the first electrode by the number of total winding turns $n_{\text{sub.1}}$, k being a natural number of 1 to $n_{\text{sub.1}}$, and wherein a ratio between a length of a radial region including the relative radial position $R_{\text{sub.1},k}$ having 10 or more stacked layers of the second portion and a length of the first surface region is 30% or more.

3. The electrode assembly according to claim 2, wherein the ratio is 30% to 85%.

4. The electrode assembly according to claim 1, wherein the second electrode includes a third portion coated with an active material and a fourth portion at a second side and adjacent the third portion, the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, wherein a part of the fourth portion is bent in the radial direction of the electrode assembly forming a second surface region including stacked layers of the fourth portion, and wherein, in a partial region of the second surface region, a number of the stacked layers of the fourth portion is 10 or more in the second axial direction of the electrode assembly.

5. The electrode assembly according to claim 4, wherein a number of total winding turns of the second electrode is defined as $n_{\text{sub.2}}$, wherein a relative radial position $R_{\text{sub.2},k}$ is defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the second electrode by the number of total winding turns $n_{\text{sub.2}}$, k being a natural number of 1 to $n_{\text{sub.2}}$, and wherein a ratio between a length of a radial region including the relative radial position $R_{\text{sub.2},k}$ having 10 or more stacked layers of the fourth portion and a length of the second surface region is 30% or more.

6. The electrode assembly according to claim 5, wherein the ratio is 30% to 85%.

7. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position $R_{\text{sub.1},1}$ of a $1_{\text{sup.st}}$ winding turn of the first electrode to a relative radial position $R_{\text{sub.1},k^*}$ of a $k^*_{\text{sup.th}}$ winding turn of the first electrode has a smaller height than a region of the second portion from a relative radial position $R_{\text{sub.1},k^*+1}$ of a $k^*+1_{\text{sup.th}}$ winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1},\text{sup.th}}$ winding turn of the first electrode.

8. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position $R_{\text{sub.1},1}$ of $1_{\text{sup.st}}$ winding turn of the first electrode to a relative radial

position R.sub.1,k* of a k*.sup.th winding turn of the first electrode has a smaller height than the first surface region.

9. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position R.sub.1,1 of a 1.sup.st winding turn of the first electrode to a relative radial position R.sub.1,k* of a k*.sup.th winding turn includes one or more layers of the second portion not bent toward the core of the electrode assembly.

10. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position R.sub.2,1 of 1.sup.st winding turn of the second electrode to a relative radial position R.sub.2,k* of a k*.sup.th winding turn of the second electrode has a smaller height than a region of the fourth portion from a relative radial position R.sub.2,k*+1 of a k*+1.sup.th winding turn of the second electrode to a relative radial position of a n.sub.2.sup.th winding turn of the second electrode.

11. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position R.sub.2,1 of a 1.sup.st winding turn of the second electrode to a relative radial position R.sub.2,k* of a k*.sup.th winding turn of the second electrode has a smaller height than the second surface region.

12. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position R.sub.2,1 of a 1.sup.st winding turn of the second electrode to a relative radial position R.sub.2,k* of a k*.sup.th winding turn of the second electrode includes one or more layers of the fourth portion not bent toward the core of the electrode assembly.

13. The electrode assembly according to claim 4, wherein the fourth portion includes a plurality of segments, each of the plurality of segments being bendable.

14. The electrode assembly according to claim 13, wherein each of the plurality of segments has a geometric shape including a base at a bending line of each of the plurality of the segments, and wherein the geometric shape includes one or more straight lines, one or more curves, or a combination thereof.

15. The electrode assembly according to claim 14, wherein the geometric shape has a width decreasing stepwise or gradually from the base to a top of the geometric shape.

16. The electrode assembly according to claim 15, wherein a lower internal angle of the geometric shape between the base and a side of at least one of the plurality of segments intersecting the base is 60 degrees to 85 degrees.

17. The electrode assembly according to claim 16, wherein the plurality of segments has a plurality of lower internal angles, the plurality of lower internal angles increasing stepwise or gradually along a direction parallel to a winding direction of the electrode assembly.

18. The electrode assembly according to claim 14, wherein each of the plurality of segments has a geometric shape having a base at a bending line of each of the plurality of segments, and wherein a lower internal angle $\theta_{\text{sub.assumption}}$ and a lower internal angle $\theta_{\text{sub.real}}$ of each of the plurality of segments satisfies the following formula:

$\theta_{\text{sub.real}} > \theta_{\text{sub.assumption}}$; and

$\theta_{\text{sub.assumption}} = 90^\circ - 360^\circ * (L_{\text{sub.arc}} / 2\pi r) * 0.5$, wherein $\theta_{\text{sub.assumption}}$ is a lower internal angle of each of the plurality of segments at a winding turn based on adjacent sides of a pair of the plurality of segments being parallel, wherein $\theta_{\text{sub.real}}$ is an actual lower internal angle at the winding turn, wherein r is a radius of the winding turn from a core center of the electrode assembly to each of the plurality of segments at the winding turn, and wherein $L_{\text{sub.arc}}$ is an arc length corresponding to a lower portion of each of the plurality of segments at the winding turn.

19. The electrode assembly according to claim 18, wherein a circumferential angle corresponding to the $L_{\text{sub.arc}}$ is 45 degrees or less.

20. The electrode assembly according to claim 18, wherein an overlapping ratio of the adjacent sides of the pair of the plurality of segments at the winding turn satisfies the following formula: $\theta_{\text{sub.real}} / \theta_{\text{sub.assumption}} - 1$, wherein the overlapping ratio of the pair of the plurality of

segments is greater than 0 and less than 0.05.

21. The electrode assembly according to claim 14, wherein a virtual circle passing through an adjacent pair of the plurality of segments at a winding turn having a radius from a core center of the electrode includes a pair of arcs passing through each segment of the adjacent pair of the plurality of segments overlapping with each other.

22. The electrode assembly according to claim 21, wherein a ratio between a length of an overlapping arc of the pair of arcs to a length of each arc of the pair of arcs passing through each of the pair of the plurality segments is defined as an overlapping ratio, and wherein the overlapping ratio is greater than 0 and less than 0.05.

23. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position $R_{\text{sub.1,1}}$ of a 1st winding turn of the first electrode to a relative radial position $R_{\text{sub.1,k}^*}$ of a k^{th} winding turn of the first electrode has a smaller height than a region of the second portion from a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^{\text{th}}+1$ winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1}}$ winding turn of the first electrode, and wherein the region from the relative radial position $R_{\text{sub.1,1}}$ of the 1st winding turn to the relative radial position $R_{\text{sub.1,k}^*}$ of the k^{th} winding turn is not bent toward the core.

24. The electrode assembly according to claim 23, wherein a length of the first electrode corresponding to the region from the relative radial position $R_{\text{sub.1,1}}$ to the relative radial position $R_{\text{sub.1,k}^*}$ is 1% to 30% compared to a length of the first electrode corresponding to the region from the relative radial position $R_{\text{sub.1,k}^*+1}$ to the relative radial position of the $n_{\text{sub.1}}$ winding turn of the first electrode.

25. The electrode assembly according to claim 2, wherein a bending length $fd_{\text{sub.1,k}^*+1}$ of the second portion at a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^{\text{th}}+1$ winding turn of the first electrode is shorter than a radial length from a relative radial position $R_{\text{sub.1,1}}$ of 1st winding turn of the first electrode to a relative radial position $R_{\text{sub.1,k}^*}$ of a k^{th} winding turn of the first electrode.

26. The electrode assembly according to claim 2, wherein a radius of the core of the electrode assembly is defined as $r_{\text{sub.c}}$, and wherein a region from a center of the core to $0.90r_{\text{sub.c}}$ is not blocked by a bent portion of the second portion located in a region from a relative radial position $R_{\text{sub.1,k}^*+1}$ of a $k^{\text{th}}+1$ winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1}}$ winding turn of the first electrode.

27. The electrode assembly according to claim 26, wherein a bending length $fd_{\text{sub.1,k}^*+1}$ of the second portion at the relative radial position $R_{\text{sub.1,k}^*+1}$ of the $k^{\text{th}}+1$ winding turn, the radius $r_{\text{sub.c}}$ of the core, and a distance $d_{\text{sub.1,k}^*+1}$ from a center of the electrode assembly to the relative radial position $R_{\text{sub.1,k}^*+1}$ satisfy the following formula:

$$fd_{\text{sub.1,k}^*+1} + 0.90 * r_{\text{sub.c}} \leq d_{\text{sub.1,k}^*+1}.$$

28. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position $R_{\text{sub.2,1}}$ of a 1st winding turn of the second electrode to a relative radial position $R_{\text{sub.2,k}^*}$ of a k^{th} winding turn has a smaller height than a region of the fourth portion from a relative radial position $R_{\text{sub.2,k}^*+1}$ of a $k^{\text{th}}+1$ winding turn of the second electrode to a relative radial position of $n_{\text{sub.2}}$ winding turn of the second electrode, and wherein the region from the relative radial position $R_{\text{sub.2,1}}$ of the 1st winding turn to the relative radial position $R_{\text{sub.2,k}^*}$ of a k^{th} winding turn is not bent toward the core.

29. The electrode assembly according to claim 28, wherein a length of the second electrode corresponding to the region from the relative radial position $R_{\text{sub.2,1}}$ to the relative radial position $R_{\text{sub.2,k}^*}$ is 1% to 30% compared to a length of the second electrode corresponding to the region from the relative radial position $R_{\text{sub.2,k}^*+1}$ to the relative radial position of $n_{\text{sub.2}}$ winding turn of the second electrode.

30. The electrode assembly according to claim 5, wherein a bending length $fd_{\text{sub.2,k}^*+1}$ of the

fourth portion at a relative radial position $R_{\text{sub.2},k^{*}+1}$ of a $k^{*}+1$.sup.th winding turn of the second electrode is shorter than a radial length from a relative radial position $R_{\text{sub.2},1}$ of a 1.sup.st winding turn of the second electrode to a relative radial position $R_{\text{sub.1},k^{*}}$ of a k^{*} .sup.th winding turn of the second electrode.

31. The electrode assembly according to claim 5, wherein a radius of the core of the electrode assembly is defined as $r_{\text{sub.c}}$, and wherein a region from a center of the core to $0.90r_{\text{sub.c}}$ is not blocked by a bent portion of the fourth portion of the second electrode located in a region from a relative radial position $R_{\text{sub.2},k^{*}+1}$ of a $k^{*}+1$.sup.th winding turn of the second electrode to a relative radial position of a $n_{\text{sub.2}}$.sup.th winding turn of the first electrode.

32. The electrode assembly according to claim 31, wherein a bending length $fd_{\text{sub.2},k^{*}+1}$ of the fourth portion at the relative radial position $R_{\text{sub.2},k^{*}+1}$ of a $k^{*}+1$.sup.th winding turn, the radius $r_{\text{sub.c}}$ of the core, and a distance $d_{\text{sub.2},k^{*}+1}$ from a center of the electrode assembly to the relative radial position $R_{\text{sub.2},k^{*}+1}$ satisfy the following formula:

$$fd_{\text{sub.2},k^{*}+1} + 0.90 * r_{\text{sub.c}} \leq d_{\text{sub.2},k^{*}+1}.$$

33. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position $R_{\text{sub.1},k^{*}+1}$ of a $k^{*}+1$.sup.th winding turn of the first electrode to a relative radial position $R_{\text{sub.1},k@}$ of a $k@$.sup.th winding turn of the first electrode is divided into the plurality of segments, heights of the plurality of segments increasing gradually or stepwise along a direction parallel to a winding direction of the first electrode.

34. The electrode assembly according to claim 33, wherein a radial length of the region from the relative radial position $R_{\text{sub.1},k^{*}+1}$ to the relative radial position $R_{\text{sub.1},k@}$ is 1% to 56% compared to a radial length of a region from a relative radial position $R_{\text{sub.1},1}$ of a 1.sup.st winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1}}$.sup.th turn of the first electrode.

35. The electrode assembly according to claim 2, wherein a region of the second portion from a relative radial position $R_{\text{sub.1},k@+1}$ of a $k@+1$.sup.th winding turn of the first electrode to a relative radial position of a $n_{\text{sub.1}}$.sup.th turn of the first electrode is divided the plurality of segments, the plurality of segments having substantially a same height from the relative radial position $R_{\text{sub.1},k@+1}$ to the relative radial position of the $n_{\text{sub.1}}$.sup.th turn of the first electrode.

36. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position $R_{\text{sub.2},k^{*}+1}$ of a $k^{*}+1$.sup.th winding turn of the second electrode to a relative radial position $R_{\text{sub.2},k@}$ of a $k@$.sup.th winding turn of the second electrode is divided into the plurality of segments, heights of the plurality of segments increasing stepwise or gradually along a direction parallel to a winding direction of the second electrode.

37. The electrode assembly according to claim 36, wherein a radial length of the region from the relative radial position $R_{\text{sub.2},k^{*}+1}$ to the relative radial position $R_{\text{sub.2},k@}$ is 1% to 56% compared to a radial length of a region from a relative radial position $R_{\text{sub.2},1}$ of a 1.sup.st winding turn of the second electrode to a relative radial position of a $n_{\text{sub.2}}$.sup.th turn of the second electrode.

38. The electrode assembly according to claim 5, wherein a region of the fourth portion from a relative radial position $R_{\text{sub.2},k@+1}$ of a $k@+1$.sup.th winding turn of the second electrode to a relative radial position of a $n_{\text{sub.2}}$.sup.th turn of the second electrode is divided into the plurality of segments, the plurality of segments having substantially a same height from the relative radial position $R_{\text{sub.2},k@+1}$ to the relative radial position of the $n_{\text{sub.2}}$.sup.th turn of the second electrode.

39. The electrode assembly according to claim 1, wherein, a region of the second portion is divided into the plurality of segments that are independently bendable, and wherein one or more heights of the plurality of segments in the first axial direction or one or more widths of the plurality of segments in a winding direction of the first electrode increases gradually or stepwise along a direction parallel to the winding direction of the first electrode individually or in groups.

40. The electrode assembly according to claim 4, wherein a region of the fourth portion is divided into a plurality of segments that are independently bendable, and wherein one or more heights of the plurality of segments in the second axial direction or one or more widths of the plurality of segments in a winding direction of the second electrode increases gradually or stepwise along a direction parallel to the winding direction of the second electrode individually or in groups.
41. The electrode assembly according to claim 13, wherein each of the plurality of segments of the second portion or the fourth portion has a width condition of 1 mm to 11 mm in the radial direction, a height condition of 2 mm to 10 mm in the first axial direction or the second axial direction, or a separation pitch condition of 0.05 mm to 1 mm in the winding direction.
42. The electrode assembly according to claim 13, further comprising: a cut groove between the plurality of segments of the second portion or the fourth portion; and a predetermined gap between a bottom of the cut groove.
43. The electrode assembly according to claim 42, wherein the predetermined gap has a length of 0.2 mm to 4 mm.
44. The electrode assembly according to claim 13, wherein the plurality of segments of the second portion or the fourth portion includes a plurality of segment groups along a winding direction of the first electrode or the second electrode, and wherein segments belonging to a same segment group of the plurality of segment groups have a substantially the same width in the winding direction, height in the first axial direction or the second axial direction, or separation pitch in the winding direction.
45. The electrode assembly according to claim 44, wherein the segments belonging to the same segment group are formed with the width in the winding direction, the height in the first axial direction or the second axial direction, or the separation pitch in the winding direction increasing gradually or stepwise along a direction parallel to the winding direction of the first electrode or the second electrode.
46. The electrode assembly according to claim 44, wherein at least a part of the plurality of segment groups are disposed at a same winding turn of the electrode assembly.
47. The electrode assembly according to claim 1, wherein the first surface region includes a stack number increasing region and a stack number uniform region from the outer circumference of the electrode assembly to the core, wherein the stack number increasing region is a region having an increasing number of the stacked layers of the second portion toward to the core of the electrode assembly, wherein the stack number uniform region is a region from a radial position where the increasing number of the stacked layers of the second portion stops to a radial position where the second portion starts to bend, and wherein a radial length of the stack number uniform region is 30% or more compared to a radial length from a winding turn where the second portion starts bending to a winding turn where the second portion stops bending.
48. The electrode assembly according to claim 5, wherein the second surface region includes a stack number increasing region and a stack number uniform region from the outer circumference of the electrode assembly to the core, wherein the stack number increasing region is a region having an increasing number of stacked layers of the fourth portion toward to the core of the electrode assembly, wherein the stack number uniform region is a region from a radial position where the increasing number of the stacked layers of the fourth portion stops to a radial position where a number of stacked layers of the fourth portion starts to bend, and wherein a radial length of the stack number uniform region is 30% or more compared to a radial length from a winding turn where the fourth portion starts bending to a winding turn where the fourth portion stops bending.
49. The electrode assembly according to claim 4, wherein the first electrode and the second electrode have a thickness of 80 μm to 250 μm , and wherein an interval of the second portion and the fourth portion at adjacent winding turns in the radial direction of the electrode assembly is 200 μm to 500 μm .
50. The electrode assembly according to claim 1, wherein the second portion of the first electrode has a thickness of 10 μm to 25 μm .

51. The electrode assembly according to claim 4, wherein the fourth portion of the second electrode has a thickness of 5 μm to 20 μm .
52. The electrode assembly according to claim 1, wherein, in the partial region of the first surface region, a total stack thickness of the stacked layers of the second portion is 100 μm to 975 μm .
53. The electrode assembly according to claim 52, wherein the second portion includes the plurality of segments that are independently bendable, wherein the first electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and wherein a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segments in the height uniform region is 1.0% to 16.3%.
54. The electrode assembly according to claim 4, wherein, in the partial region of the second surface region, a total stack thickness of stacked layers of the fourth portion is 50 μm to 780 μm .
55. The electrode assembly according to claim 54, wherein the fourth portion includes a plurality of segments that are independently bendable, wherein the second electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and wherein a ratio of a stack thickness of the fourth portion of the second surface region to the substantially uniform height of the plurality of segments in the height uniform region is 0.5% to 13.0%.
56. An electrode assembly, comprising: a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region, wherein, in a partial region of the first surface region, a stack thickness of the second portion is 100 μm to 975 μm , wherein the second portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly.
57. The electrode assembly according to claim 56, wherein the second portion of the first electrode includes the plurality of segments that are independently bendable, wherein the first electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region with the plurality of segments having a substantially uniform height, and wherein a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segments in the height uniform region is 1.0% to 16.3%.
58. An electrode assembly, comprising: a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein the second electrode includes a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, wherein a part of the fourth portion is bent in a radial direction of the electrode assembly forming a second surface region, wherein, in a partial region of the second surface region, a stack thickness of the fourth portion is 50 μm to 780 μm , wherein the fourth portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly.

59. The electrode assembly according to claim 58, wherein the fourth portion includes a plurality of segments that are independently bendable, wherein the second electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and wherein a ratio of a stack thickness of the fourth portion of the second surface region to the substantially uniform height of the segment is 0.5% to 13.0%.

60. A battery, comprising: an electrode assembly comprising: a first electrode; a second electrode; and a separator between the first electrode and the second electrode, the first electrode, the second electrode, and the separator wound about an axis defining a core and an outer circumference, wherein at least one of the first electrode or the second electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein at least a part of the second portion is bent in a radial direction of the electrode assembly forming a first surface region including stacked layers of the second portion, wherein, in a partial region of the first surface region, a number of the stacked layers of the second portion is 10 or more, wherein the second portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly; a battery housing accommodating the electrode assembly and being electrically connected to one of the first electrode or the second electrode to have a first polarity; a sealing body sealing a first opening of the battery housing; a terminal electrically connected to the other one of the first electrode or the second electrode to have a second polarity, the terminal having a surface exposed to an outside of the battery housing; and a current collector welded to the first surface region and electrically connected to one of the battery housing or the terminal, wherein a welding region of the current collector overlaps with the first surface region having 10 or more of the stacked layers of the second portion.

61. The battery according to claim 60, wherein a number of total winding turns of the first electrode is defined as $n_{\text{sub.1}}$, wherein a relative radial position $R_{\text{sub.1},k}$ is defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the first electrode by the number of total winding turns $n_{\text{sub.1}}$, k being a natural number of 1 to $n_{\text{sub.1}}$, and wherein a ratio between a length of a radial region including the relative radial position $R_{\text{sub.1},k}$ having 10 or more stacked layers of the second portion and a length of the first surface region is 30% or more.

62. The battery according to claim 60, wherein the second electrode includes a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion being exposed beyond the separator along a second axis direction of the electrode assembly, wherein a number of total winding turns of the second electrode is defined as $n_{\text{sub.2}}$, wherein a relative radial position $R_{\text{sub.2},k}$ is defined by dividing a winding turn index k at a $k_{\text{sup.th}}$ winding turn location of the first electrode by the number of total winding turns $n_{\text{sub.2}}$, k being a natural number of 1 to $n_{\text{sub.2}}$, and wherein a ratio between a length of a radial region including the relative radial position $R_{\text{sub.2},k}$ having 10 or more stacked layers of the fourth portion and a length of the second surface region is 30% or more.

63. The battery according to claim 60, wherein 50% or more of the welding region of the current collector overlaps with the first surface region having 10 or more stacked layers of the second portion.

64. The battery according to the claim 63, wherein a welding strength of the current collector is 2 kgf/cm^{sup.2} or more.

65. A battery, comprising: an electrode assembly including a first electrode, a second electrode, and a separator between the first electrode and the second electrode, wherein the first electrode, the second electrode, and the separator are wound about an axis to define a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed

beyond the separator along a first axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly to form a first surface region, wherein, in a partial region of the first surface region, a stack thickness of the second portion is 100 μm to 975 μm , wherein the second portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly; a battery housing accommodating the electrode assembly and electrically connected to one of the first electrode or the second electrode to have a first polarity; a sealing body sealing a first end of the battery housing; a terminal electrically connected to the other of the first electrode or the second electrode to have a second polarity and configured to have a surface exposed to the outside of the battery housing; and a first current collector welded to the first surface region and electrically connected to one of the battery housing or the terminal, wherein a welding region of the first current collector overlaps with the partial region of the first surface region having the stack thickness of the second portion is 100 μm to 975 μm .

66. The battery according to claim 65, wherein the second portion of the first electrode is divided into the plurality of segments that are independently bendable, wherein the first electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and wherein a ratio of a stack thickness of the second portion in the first surface region to the substantially uniform height of the plurality of segment in the height uniform is 1.0% to 16.3%.

67. The battery according to the claim 65, wherein a welding strength of the first current collector is 2 kgf/cm² or more.

68. A battery, comprising: an electrode assembly including a first electrode, a second electrode, and a separator between the first electrode and the second electrode, wherein the first electrode, the second electrode, and the separator are wound about an axis to define a core and an outer circumference, wherein the first electrode includes a first portion coated with an active material and a second portion at a first side and adjacent to the first portion, the second portion being exposed beyond the separator along a first axial direction of the electrode assembly, wherein the second electrode includes a third portion coated with an active material and a fourth portion at a second side and adjacent to the third portion, and the fourth portion being exposed beyond the separator along a second axial direction of the electrode assembly, wherein a part of the second portion is bent in a radial direction of the electrode assembly to form a first surface region, wherein a part of the fourth portion is bent in the radial direction of the electrode assembly forming a second surface region, wherein, in a partial region of the second surface region, a stack thickness of the fourth portion is 50 μm to 780 μm , wherein the fourth portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly; a battery housing accommodating the electrode assembly and electrically connected to one of the first electrode or the second electrode to have a first polarity; a sealing body sealing a first end of the battery housing; a terminal electrically connected to the other of the first electrode or the second electrode to have a second polarity and configured to have a surface exposed to the outside of the battery housing; and a second current collector welded to the second surface region and electrically connected to one of the battery housing or the terminal, wherein a welding region of the second current collector overlaps with the partial region of the second surface region in which the stack thickness of the fourth portion is 50 μm to 780 μm .

69. The battery according to claim 68, wherein the fourth portion of the second electrode is divided into the plurality of segments that are independently bendable, wherein the second electrode includes a height variable region having the plurality of segments with variable heights and a height uniform region having the plurality of segments with a substantially uniform height, and wherein a ratio of a stack thickness of the fourth portion in the second surface region to the

substantially uniform height of the plurality of segments in the height uniform region is 0.5% to 13.0%.

70. The battery according to the claim 68, wherein a welding strength of the second current collector is 2 kgf/cm.^{sup.2} or more.

71. The battery according to claim 68, further comprising a first current collector welded to the first surface region and electrically connected to the other of the battery housing or the terminal, wherein the welding region of the first current collector overlaps with the partial region of the first surface region in which the stack thickness of the second portion is 100 μm to 975 μm by 50% or more.

72. The battery according to claim 68, wherein the welding region of the second current collector overlaps with the partial region of the second bending surface region in which the stack thickness of the fourth portion is 50 μm to 780 μm by 50% or more.

73. A battery pack, comprising the battery according to claim 60.

74. A vehicle, comprising the battery pack according to claim 73.

75. A battery pack, comprising the battery according to claim 65.

76. A vehicle, comprising the battery pack according to claim 75.

77. The electrode assembly according to claim 1, wherein a ratio between a length of a radial region having 10 or more stacked layers of the second portion and a length of the first surface region is 30% or more.

78. A method of making an electrode assembly for a battery, comprising: providing a first electrode; providing a second electrode; providing a separator between the first electrode and the second electrode, winding the first electrode, the second electrode, and the separator about an axis to define a core and an outer circumference of the electrode assembly; coating a first portion on the first electrode with an active material; exposing a second portion on the first electrode beyond the separator along a first axial direction of the electrode assembly; bending a part of the second portion in a radial direction of the electrode assembly to form a first surface region including stacked layers of the second portion; and forming in a partial region of the first surface region, 10 or more stacked layers of the second portion in the first axial direction, wherein the second portion includes a plurality of segments, and wherein segments immediately adjacent each other in the radial direction of the electrode assembly are offset in a circumferential direction of the electrode assembly.

79. The method of claim 78, further comprising: coating a third portion on the second electrode with the active material; exposing a fourth portion on the second electrode beyond the separator along a second axial direction of the electrode assembly; bending a part of the fourth portion in a radial direction of the electrode assembly to form a second surface region including stacked layers of the fourth portion; and forming in a partial region of the second surface region, 10 or more stacked layers of the fourth portion in the second axial direction.
